TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS

STUDY OF THE SURFACE HEATING EFFECT ON A COMPRESSIBLE WALL JET USING RAINBOW SCHLIEREN DEFLECTOMETRY

By:
Amogh Adhikari (078BAS006)
Anup Bajgain (078BAS007)
Toran Bahadur Niroula (078BAS043)
Yogesh Dhami (078BAS046)

A FINAL REPORT

SUBMITTED TO THE DEPARTMENT OF MECHANICAL AND AEROSPACE ENGINEERING IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR IN AEROSPACE ENGINEERING

DEPARTMENT OF MECHANICAL AND AEROSPACE ENGINEERING
LALITPUR, NEPAL

June 2026

COPYRIGHT

The authors have agreed that the Library, Department of Mechanical and Aerospace Engineering, Institute of Engineering, Pulchowk Campus, may make this report freely available for inspection. Moreover, the authors have agreed that permission for extensive copying of this project report for scholarly purposes may be granted by the supervisors who supervised the project work recorded herein or, in their absence, by the Head of the Department wherein the project report was done. It is understood that recognition will be given to the authors of this project report and to the Department of Mechanical and Aerospace Engineering, Pulchowk Campus, Institute of Engineering, in any use of the material of this project report. Copying or publication or the other use of this report for financial gain without approval of the Department of Mechanical and Aerospace Engineering, Institute of Engineering, Pulchowk Campus and authors' written permission is strictly prohibited.

Request for permission to copy or to make any other use of this project report in whole or in part should be addressed to:

Head of Department
Department of Mechanical and Aerospace Engineering,
Institute of Engineering, Pulchowk Campus,
Lalitpur, Nepal

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

DEPARTMENT OF MECHANICAL AND AEROSPACE ENGINEERING

LETTER OF APPROVAL

The undersigned certify that they have read, and recommended to the Institute of Engineering for acceptance, a project report entitled **'STUDY OF THE SURFACE HEATING EFFECT ON A COMPRESSIBLE WALL JET USING RAINBOW SCHLIEREN DEFLECTOMETRY'** submitted by **Amogh Adhikari, Anup Bajgain, Toran B. Niroula and Yogesh Dhami** in partial fulfillment of the requirements for the Bachelor's Degree in Aerospace Engineering.

Supervisor : **Kamal Darlami,** Assistant Professor
Department of Mechanical and Aerospace Engineering
Institute of Engineering, Pulchowk Campus

Supervisor : **Laxman Motra,** Assistant Professor
Department of Mechanical and Aerospace Engineering
Institute of Engineering, Pulchowk Campus

Akin

External Examiner: **Er. Akin Chhetri,** CFD engineer
SG design

Head of Department : **Dr. Sudip Bhattrai,** Assistant Professor
Department of Mechanical and Aerospace Engineering
Institute of Engineering, Pulchowk Campus

DATE OF APPROVAL : 2083/01/30

ABSTRACT

This project investigates a high-subsonic compressible wall jet flowing over a temperature-controlled flat plate using a combined experimental and computational approach. A 3D-printed converging nozzle (12 mm $\times$ 5 mm exit) generates a tangential wall jet at nozzle pressure ratios of approximately 1.5 and 1.68, corresponding to estimated exit Mach numbers of $M \approx 0.75$ and $M \approx 0.90$, over a Peltier-heated plate maintained at surface temperatures of 40 and 60°C. The experiment is conducted at $p_\infty \approx 0.871$ bar.

Rainbow schlieren deflectometry (RSD) is employed as the primary quantitative diagnostic. To process the resulting image sequences in a reproducible and documented manner, a dedicated Python library, *pyrsd* (v0.0.1), was developed as part of this work. It implements filter calibration, displacement extraction, ensemble statistics, and density-field reconstruction via both one-dimensional cumulative integration and two-dimensional Poisson inversion. Statistical convergence of the mean is verified by computing partial means at increasing frame counts and confirming stabilization before the full dataset is used. The RSD pipeline is validated against natural convection from the heated plate. Mean density-gradient fields, density and temperature fields of the displacement sequences characterize the effect of wall heating on near wall density structure. Complementary 2D steady RANS simulations in ANSYS Fluent provide density-gradient fields for qualitative comparison.

Keywords: *rainbow schlieren deflectometry; compressible wall jet; density gradient; heated flat plate; pyrsd*

ACKNOWLEDGEMENT

First and foremost, we would like to express our sincere appreciation to our project supervisors **Asst. Prof. Kamal Darlami** and **Asst. Prof. Laxman Motra** for their supervision. Constant guidance, encouragement, and critical feedback from **Asst. Prof. Kamal Darlami** has always been uplifting to us. His valuable insights on this project and expertise on Schlieren Imaging and Flow Visualization in general has been immensely instrumental to the success of this project. We are also grateful to **Asst. Prof. Laxman Motra** for his suggestions, and feedbacks which consistently kept our work on the right track.

We are also thankful to **Asst. Prof. Dr. Sudip Bhattrai** for his insightful feedbacks on experimental setup and procedures. We are grateful to **Asst. Prof. Biman Rimal** for helping us procure tools and instruments for the project. We would like to thank **Salim Maharjan** for sharing his knowledge and skills generously.

We extend our gratitude to **Flow Visualization Laboratory, Pulchowk Campus** and the **Department of Mechanical and Aerospace Engineering** for providing us workspace, tools, and instruments as well as the opportunity to learn, grow, and explore.

We also extend our thank to all our teachers, colleagues, seniors, and juniors for their direct and indirect support during the project. Finally, we give a deep sense of appreciation to our family members who have stood with us throughout this journey.

We appreciate and acknowledge any kind of suggestions or constructive criticisms that may have improved the progress and quality of our work.

Authors:
Amogh Adhikari
Anup Bajgain
Toran Bahadur Niroula
Yogesh Dhami

Contents

COPYRIGHT

LETTER OF APPROVAL

ABSTRACT **i**

ACKNOWLEDGEMENT **ii**

TABLE OF CONTENTS **iii**

LIST OF TABLES **vi**

LIST OF FIGURES **viii**

LIST OF ABBREVIATIONS **ix**

LIST OF SYMBOLS **x**

1 INTRODUCTION **1**

1.1 Background 1

1.1.1 Rainbow Schlieren Deflectometry 2

1.1.2 Wall Jet Flows 3

1.1.3 Compressibility and Thermal Effects 3

1.2 Problem Statement 4

1.3 Objectives 4

1.3.1 Main Objective 4

1.3.2 Specific Objectives 5

1.4 System Requirements 6

2 LITERATURE REVIEW **8**

2.1 Wall Jets 8

2.2 Schlieren Imaging 9

2.3 Rainbow Schlieren Deflectometry 9

2.4 Density Field Reconstruction from Gradient Data 10

2.5 Natural Convection as a Validation Case 11

2.6 Research Gaps and Motivation 11

3 METHODOLOGY **12**

3.1 Experimental Setup 14
3.1.1 Compressed Air Supply and Pressure Estimation 14
3.1.2 Nozzle Design 15
3.1.3 Heated Plate Assembly 16
3.2 Schlieren System 20
3.2.1 Aperture design and fabrication 20
3.2.2 Filter Design, Generation, and Fabrication 21
3.2.3 Filter Selection for Quantitative Measurements 23
3.2.4 Conventional (Knife-Edge) Schlieren 24
3.2.5 Rainbow Schlieren Deflectometry - Calibration 24
3.2.6 RSD Measurement Principles 25
3.3 pyrsd: RSD Processing Library 28
3.4 RSD Processing Pipeline 29
3.4.1 Ensemble Convergence Verification 30
3.5 CFD Methodology 30
3.5.1 Mesh Independence Test 31
3.5.2 Plumbing Simulation 33
3.5.3 Wall-jet Simulation 34
3.6 Test Matrix 36
3.7 Sources of Uncertainty 36

4 RESULTS AND DISCUSSION **37**

4.1 Schlieren System Validation and Filter Characterisation 37
4.2 Nozzle Jet Characterization 38
4.2.1 Ensemble Convergence 38
4.3 Natural Convection Validation 39
4.3.1 Raw Hue Field and Mean Density Gradient 39
4.3.2 Reconstructed Density and Temperature Fields 41
4.3.3 Wall Normal Profiles and Wall Temperature Validation 42
4.4 Wall Jet Results 43
4.4.1 Mean Displacement Field 43
4.4.2 Mean Density Gradient Field 45
4.4.3 Wall Normal Gradient Profiles 46
4.4.4 Effect of Wall Temperature 47
4.4.5 Effect of Reservoir Pressure 50
4.4.6 Streamwise Decay of Peak Density Gradient 52

4.5 CFD Results and Comparison with Experiment 53
4.6 Qualitative RSD Flow Visualisation 57
4.7 Limitations 61
4.8 Problems Encountered 61
4.9 Budget Analysis 62

5 CONCLUSION AND SCOPE OF FURTHER ENHANCEMENT 63
5.1 Conclusions 63
5.2 Scope for Further Enhancement 64

REFERENCES 65

A Nozzle Design: Quasi-One-Dimensional Analysis 68

LIST OF TABLES

Table 1.1 Hardware requirements. 6
Table 1.2 Software requirements. 7

Table 3.1 Estimated nozzle exit conditions ($T_\infty = 289$ K, $p_\infty = 0.871$ bar, supply-line loss factor ≈ 0.70 from CFD of plumbing). 15
Table 3.2 Comprehensive System Wiring Connections 17
Table 3.3 *pyrsd* principal functions. 28
Table 3.4 Mesh independence test results for the plumbing assembly . . . 32
Table 3.5 Mesh independence test results for the wall-jet configuration. . . 32
Table 3.6 Boundary conditions for the 2-D wall-jet simulation 35
Table 3.7 Wall jet test matrix. 36
Table 3.8 Measurement uncertainty sources. 36

Table 4.1 Budget analysis. 62

LIST OF FIGURES

Figure 1.1 Schematic diagram of free jet growth [6]. 2
Figure 1.2 Schematic diagram of a wall jet over a flat plate.[11] 3

Figure 3.1 Overall methodology of the present investigation. 12
Figure 3.2 experimental and post processing work flow. 13
Figure 3.3 Experimental facility. The plate and nozzle assembly. 14
Figure 3.4 3D-printed converging nozzle (exit 12 mm × 5 mm, contraction ratio 9.25). 16
Figure 3.5 Schematic diagram of PID system 18
Figure 3.6 Peltier Module and thermistor assembly in the metal place. . . . 18
Figure 3.7 Response of PID system for 60°C 19
Figure 3.8 PWM response to the temperature. 19
Figure 3.9 Complete experimental setup with Z-type double-mirror schlieren system. 20
Figure 3.10 Rectangular source aperture used in RSD 21
Figure 3.11 The three slides with filter of varying widths and hue ranges . . 23
Figure 3.12 RSD calibration curve: hue versus filter displacement. 24
Figure 3.13 Meshing of plumbing assembly 33
Figure 3.14 Meshing of 2-D domain 34

Figure 4.1 Schlieren images illustrating the effect of filter fabrication medium. 37
Figure 4.2 Rolling coefficient of variation (RCoV) of the mean absolute displacement. 39
Figure 4.3 Hue extracted RSD snapshots of the natural convection boundary layer. 40
Figure 4.4 Mean density gradient magnitude for natural convection from the heated vertical plate. 40
Figure 4.5 mean density and temperature fields reconstructed from the RSD measurement at $T_w = 40°C$. 41
Figure 4.6 mean density and temperature fields reconstructed from the RSD measurement at $T_w = 60°C$. 41
Figure 4.7 Wall normal density and temperature profiles at $y = 3$, 11, and 18 mm for $T_w = 40°C$. 42

Figure 4.8 Wall normal density and temperature profiles at $y = 3$, 11, and 18 mm for $T_w = 60°$C. 42
Figure 4.9 mean displacement fields for the wall jet at 1.0 bar gauge. 44
Figure 4.10 mean displacement fields for the wall jet at 1.4 bar gauge. 44
Figure 4.11 mean density gradient fields for the wall jet at $T_w = 40°$C. . . . 45
Figure 4.12 mean density gradient fields for the wall jet at $T_w = 60°$C. . . . 46
Figure 4.13 Wall normal normalized density-gradient profiles at four streamwise stations. 47
Figure 4.14 Effect of wall temperature on the wall normal density gradient profiles at two streamwise stations and two operating pressures. Each plot shows $|\partial\rho/\partial y|$ versus wall normal distance y. 49
Figure 4.15 Effect of reservoir pressure on the wall normal density gradient profiles at two streamwise stations and two wall temperatures. . 51
Figure 4.16 Streamwise decay of the peak mean density gradient at $T_w =$ 40°C for both operating pressures. 52
Figure 4.17 Streamwise decay of the peak mean density gradient at $T_w =$ 60°C for both operating pressures. The elevated wall temperature increases peak gradient magnitudes at both pressures. 53
Figure 4.18 Comparison of wall normal density gradient profiles: experimental RSD and 2D RANS CFD at four streamwise stations ($p_{\text{gauge}} = 1.0$ bar, $T_w = 60°$C).. 56
Figure 4.19 RSD image of a candle plume. 57
Figure 4.20 Comparison of knife-edge schlieren and RSD for a matchstick ignition, acquired in separate runs under identical conditions. . . 58
Figure 4.21 RSD image sequence of a candle flame descending into rising butane gas plumes, acquired at 40 fps ($\Delta t = 25$ ms between panels). The green background corresponds to the filter center hue (zero deflection). 60

Figure A.1 Engineering drawings of the 3D-printed converging nozzle. . . . 69

LIST OF ABBREVIATIONS

2D	2 Dimensional
3D	3 Dimensional
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
CFL	Courant-Friedrichs-Lewy
DAQ	Data Acquisition
DMAE	Department of Mechanical and Aerospace Engineering
FPS	Frames Per Second
GCI	Grid Convergence Index
HP	Horse Power
IR	Infrared
IRT	Infrared Thermography
LED	Light Emitting Diode
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
NPR	Nozzle Pressure Ratio
NRs	Nepalese Rupees
PID	Proportional-Integral-Derivative
RANS	Reynolds-Averaged Navier-Stokes
RSD	Rainbow Schlieren Deflectometry
ROI	Region of Interest
SNR	Signal-to-Noise Ratio
TBL	Turbulent Boundary Layer
k-ε	k-epsilon turbulence model
k-ω **SST**	k-omega Shear Stress Transport turbulence model

LIST OF SYMBOLS

Symbol	Description
$A(z)$	Local nozzle cross-sectional area
A^*	Critical area at $M = 1$
d	Nozzle width (rectangular)
Δd	RSD displacement field (filter plane displacement)
E	Total energy per unit mass;
e	Internal energy per unit mass
f_2	Focal length of decollimating mirror
H, W	Nozzle height and width
k	Turbulent kinetic energy; also thermal conductivity
K_{GD}	Gladstone-Dale constant
L	Nozzle length; also spanwise test-section depth
$\dot{m}$	Mass flow rate
n	Refractive index
n_0	Ambient refractive index
p	Static pressure
p_0	Total (stagnation) pressure
P_k	Production of turbulent kinetic energy
R	Specific gas constant for air
Re	Reynolds number
T	Static temperature
T_0	Total (stagnation) temperature
T_w	Wall (plate surface) temperature
T_∞	Ambient temperature
t	Time
$\mathbf{u}$	Velocity vector
V	Flow velocity magnitude
x, y, z	Cartesian coordinates
ε	deflection angle
μ	Dynamic viscosity
ν	Kinematic viscosity
ρ	Fluid density
ρ_∞	Ambient density

Chapter 1: INTRODUCTION

1.1 Background

Rainbow Schlieren Deflectometry (RSD) is a non-invasive optical measurement technique used to visualize and quantify the refractive index gradients in transparent media[1]. It evolved from conventional gray-scale schlieren methods to overcome their limitations in providing quantitative data [2]. Such optical techniques can be utilized to study fluid flow and its interactions. When a jet issues from a nozzle and impinges on or flows tangentially over a surface, strong velocity gradients, pressure variations, and density gradients develop, leading to highly unsteady flow structures and complex momentum and heat transfer mechanisms[3].

In compressible flow regimes, even at subsonic Mach numbers, variations in temperature and pressure produce measurable changes in density. These density changes are directly linked to the refractive index through the Gladstone-Dale relationship[4]. Such gradients make optical diagnostics such as schlieren and, especially, Rainbow Schlieren Deflectometry powerful tools as they provide non-invasive, high-resolution visualization by correlating the angular deflection of light with color or hue [5]. When the impinged surface is heated, the interaction becomes further coupled: thermal gradients modify local density fields, influence jet development, and alter near-wall flow behavior. Understanding this coupled interaction between compressible jets and heated surfaces is therefore essential for the design of efficient cooling systems, high-speed aerodynamic surfaces, and propulsion-related components[3].

The main principle of the RSD is based on the conversion of light deflection angles into a hue-coded map(color filter) [5]. The refraction of collimated light rays occurs while passing through a medium characterized by local density differences resulting from thermal or concentration distributions.

This project focuses on the experimental and numerical investigation of a high-subsonic jet interacting with a heated flat plate, with emphasis on flow structure, density-gradient fields, and temperature-gradient.

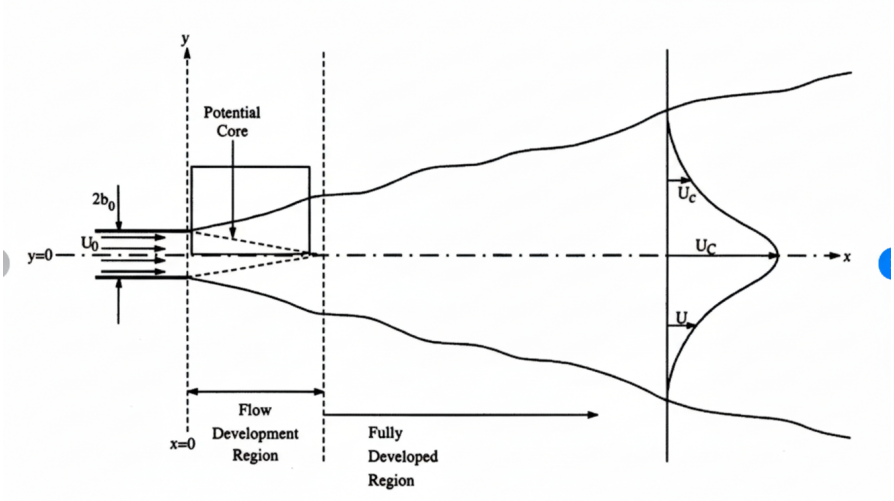


Figure 1.1: Schematic diagram of free jet growth [6].

1.1.1 Rainbow Schlieren Deflectometry

Rainbow Schlieren Deflectometry (RSD), often referred to as color schlieren, provides spatial distributions of state variables such as temperature, pressure, or concentration in transparent media[1]. Unlike conventional schlieren techniques, RSD provides refractive index gradient fields by characterizing light attenuation resulting from its refraction[5][7]. This capability allows us to relate the detected hue directly to the thermophysical property of interest through a calibrated color model, typically the Hue-Saturation-Intensity (HSI) scheme[5][4].

The deflected light rays pass at a specific cumulative angle proportional to the integrated gradient along the path[8][1]. After exiting the test section, these disturbed rays are focused by a convex lens or a parabolic mirror onto a rainbow filter positioned in the focal plane, where their displacement from the focal point can be observed as a change in color or hue[1][5][4]. By comparing the hue values of a schlieren image to an undisturbed reference image and applying a calibration curve, the angular deflection can be accurately quantified[4][1]. This relationship is fundamentally governed by the Gladstone-Dale relationship, which links the refractive index of a fluid to its density[9][5][4].

$$(n-1) = \sum C_i \rho_i \tag{1.1}$$

The constant C, called the Gladstone-Dale constant, is a function of the particular gas and varies slightly with wavelength.

1.1.2 Wall Jet Flows

When a jet exits a nozzle flush with or tangential to a surface, the flow comprises a potential core, developing shear layers on the outer (free) side, and a wall boundary layer on the inner (wall) side. Potential core is the central core of the jet that remains unaffected by the surrounding ambient fluid[10]. Surrounding the potential core, there is a developing shear layer where the fluid from the surrounding gets entrained. This causes the jet to expand radially, and its centerline velocity decreases. In Downstream of the potential core, these regions merge and the flow transitions to a self-similar wall jet. The interaction gives rise to strong velocity gradients, coherent vortical structures, and, incompressible conditions, pronounced density variations.

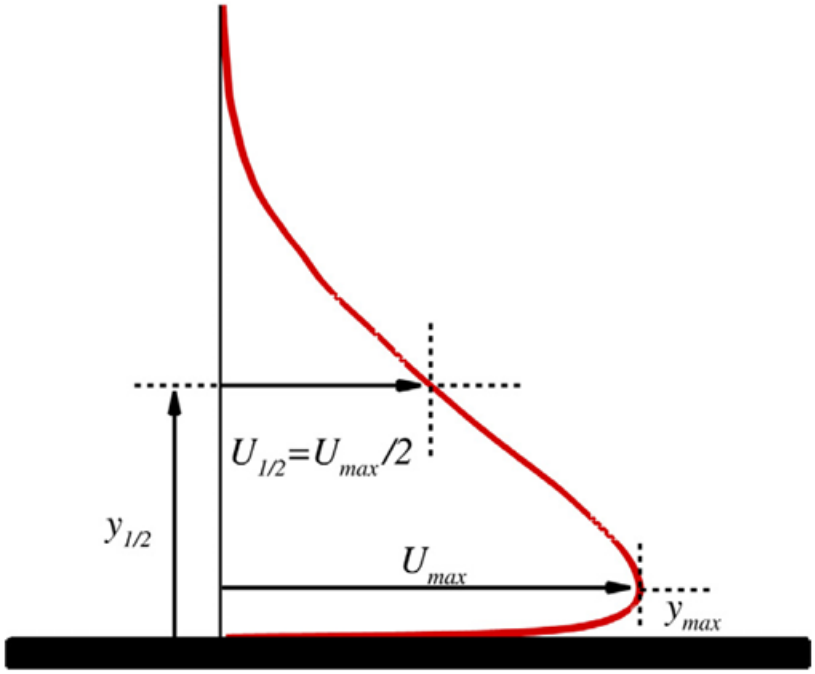


Figure 1.2: Schematic diagram of a wall jet over a flat plate.[11]

1.1.3 Compressibility and Thermal Effects

At high-subsonic Mach numbers ($M \gtrsim 0.5$), compressibility effects become non-negligible: relatively small changes in temperature and velocity produce measurable density changes accessible to schlieren-based optical diagnostics. A heated isothermal wall introduces additional density stratification in the near-wall region and modifies shear-layer development through buoyancy and viscosity changes. The present study is conducted at Kathmandu altitude (approximately 1400 m above sea level), where the ambient pressure

is $p_\infty \approx 0.871$ bar notably lower than the sea-level standard value. This reduces the critical nozzle pressure ratio for choking and must be accounted for when estimating operating conditions from regulator gauge readings.

1.2 Problem Statement

The interaction of a compressible jet with a heated surface involves complex coupling between velocity, pressure, temperature, and density fields. While wall jets have been extensively studied in the incompressible regime, comparatively few studies experimentally examine high-subsonic wall jets over heated surfaces using quantitative optical diagnostics. Furthermore, rainbow schlieren deflectometry, despite its demonstrated capability for quantitative density measurement lacks a standardised, documented processing pipeline: each research group implements calibration and reconstruction differently, reducing reproducibility and making inter-study comparison difficult.

This project addresses these gaps by:

i. building an experimental setup for a heated-wall compressible wall jet;

ii. developing *pyrsd*, an open, documented Python library implementing the complete Greenberg RSD pipeline from filter calibration to density reconstruction;

iii. validating the pipeline quantitatively against natural convection in vertical orientation.

iv. applying RSD to characterize the effect of wall heating on density-gradient structures.

1.3 Objectives

1.3.1 Main Objective

To experimentally and numerically investigate the effect of wall heating on a compressible wall jet, with emphasis on quantitative density-gradient characterization using calibrated rainbow schlieren deflectometry and complementary CFD.

1.3.2 Specific Objectives

i. To design and characterize a compressible jet facility incorporating a converging nozzle and a temperature-controlled flat plate.

ii. To implement and calibrate a Rainbow Schlieren Deflectometry system and obtain quantitative mean density-gradient, density, and temperature fields for the wall jet at selected operating conditions.

iii. To perform 2D CFD analysis and compare it with experimental data.

iv. To develop and validate *pyrsd*, a Python library for RSD

1.4 System Requirements

Hardware Requirements

Table 1.1: Hardware requirements.

S.N.	Component	Function
i.	Z-type schlieren system with color filter	Visualization and quantitative measurement of density gradients.
ii.	High-speed camera	RSD image acquisition at up to 24,000 fps.
iii.	3D printer	Fabrication of converging nozzle and mounts.
iv.	NTC thermistors	Monitoring and logging of plate surface temperature.
v.	Peltier modules + Arduino Nano PID	Controlled isothermal plate heating (40–80°C).
vi.	Pressure regulator + Bourdon gauge	Setting and reading supply gauge pressure.
vii.	Micrometer translation stage	Precise filter displacement during RSD calibration.

Software Requirements

Table 1.2: Software requirements.

S.N.	Software	Function
i.	*pyrsd* v0.0.1 (in-house)	RSD calibration, processing, density reconstruction, convergence analysis.
ii.	ANSYS Fluent	Steady compressible RANS CFD.
iii.	Python (NumPy, SciPy, Matplotlib)	Post-processing and figure generation.
iv.	SOLIDWORKS	CAD design
v.	OpenFOAM	Preliminary 3D CFD (midterm phase).

Chapter 2: LITERATURE REVIEW

The present study sits at the intersection of compressible wall-jet aerodynamics and quantitative optical flow diagnostics. Because the experimental contribution of this work centers on Rainbow Schlieren Deflectometry (RSD) and its processing pipeline, this review gives primary emphasis to the development and application of RSD and related schlieren techniques, with supporting coverage of wall-jet fluid mechanics and modal analysis.

2.1 Wall Jets

The wall jet represents a shear flow in which a jet issues tangentially along a surface, producing a flow field characterized by strong velocity gradients near the wall and a free shear layer on the outer side. Glauert [12] derived the earliest theoretical similarity solutions, identifying an inner wall-dominated region and an outer momentum-driven region, a two-region structure that remains the standard framework. Bradshaw and Gee [13] confirmed many aspects experimentally but identified near-field deviations from similarity attributed to initial conditions and nozzle geometry. Launder and Rodi [14] provided a comprehensive review emphasizing turbulence modeling challenges and the sensitivity of wall jets to exit conditions.

Most classical wall-jet studies have been conducted in the incompressible regime. At high-subsonic Mach numbers, compressibility modifies turbulence structure, suppresses mixing, and introduces measurable density gradients that directly affect optical diagnostics. This regime has received comparatively less attention. Wall heating further complicates the picture: Myers *et al.* [15] demonstrated strong coupling between turbulence intensity and thermal boundary-layer growth, while more recent work shows that Wall temperature modifies near-wall viscosity and density, altering momentum transport. The flush-mounted nozzle configuration used in the present study, in which the lower nozzle lip aligns with the plate surface, has been studied even less, as most Investigations consider offset jets or configurations with a finite nozzle-to-wall gap.

2.2 Schlieren Imaging

Schlieren techniques visualize density gradients by converting phase differences of a transmitted light beam into amplitude or color variations. In compressible and thermally stratified flows, schlieren imaging reveals shear layers, mixing regions, shock waves, expansion fans, and thermal boundary layers. Settles [16] provides the definitive reference for schlieren and shadowgraph system design, covering the Z-type double-mirror arrangement, source geometry, knife-edge cutoff orientation, sensitivity, and optical alignment. A key design consideration highlighted by Settles is the trade-off between sensitivity and field of view: a narrower source aperture or knife-edge cutoff increases sensitivity but reduces the usable dynamic range.

2.3 Rainbow Schlieren Deflectometry

Rainbow Schlieren Deflectometry replaces the knife-edge or stripe filter with a continuously varying color filter at the focal plane, enabling quantitative measurement of the ray deflection angle from the recorded hue field. The technique was systematized by Greenberg, Klimek, and Buchele [2], who introduced the calibration procedure in which the filter is displaced in known increments across the focal plane without flow, recording the hue at each step to produce a hue to displacement lookup table. This decouples the measurement from light source intensity fluctuations and is essential for quantitative analysis.

Agrawal and colleagues extended RSD significantly. Agrawal *et al.* [5] provided a detailed treatment of the complete RSD methodology, from filter design and fabrication through calibration, image processing, and density reconstruction, for axisymmetric flows using Abel inversion. Their work established key design parameters: filter width (related to the measurable deflection range), color gradient uniformity, and the importance of the source aperture geometry. A rectangular slit source, which they and subsequent works employed, provides higher illumination and greater sensitivity along one direction compared with a circular pinhole, at the cost of directional sensitivity.

For non-axisymmetric flows such as the wall jet studied here, Abel inversion is inapplicable and density must be recovered by integration of the measured gradient field. Elsinga *et al.* [7] compared RSD with background-oriented schlieren (BOS) and assessed the accuracy of density reconstruction from integrated gradient fields, highlighting that

one-dimensional integration accumulates error along the integration path while two-dimensional Poisson inversion uses both gradient components simultaneously and is globally more accurate, though sensitive to noise in the gradient field.

Miyazato and colleagues [17] applied RSD to supersonic and under-expanded jets, demonstrating its capability to resolve sharp density-gradient structures including oblique shocks and expansion fans. The presence of such structures in the present facility at higher regulator pressures (Section 3.1) makes their work directly relevant.

Settles and Hargather reviewed quantitative schlieren and BOS methods comprehensively [18], comparing sensitivity, resolution, and practical requirements. Hargather and Settles [19] used BOS to measure density fields in natural convection from a heated surface, providing a benchmark dataset against which RSD-derived results can be compared for validation purposes.

2.4 Density Field Reconstruction from Gradient Data

Given a measured density-gradient field $(\partial\rho/\partial x, \partial\rho/\partial y)$, the density field $\rho(x,y)$ can be recovered by integration. The simplest approach is one-dimensional (1D) cumulative integration along rows or columns, anchored at a reference pixel of known density (typically the ambient column in the schlieren field of view). This is straightforward to implement but the integration error accumulates monotonically along each line, producing characteristic streaks in the reconstructed field when the gradient data are noisy. a phenomenon well-documented in the literature and observed in the present work.

Two-dimensional Poisson inversion avoids this by solving $\nabla^2\rho = \nabla\cdot(\nabla\rho)$ globally, using both gradient components as the right-hand side and enforcing a single Dirichlet condition at the reference point. This minimizes the global reconstruction error but requires the right-hand side (the divergence of the gradient field, equivalent to the second spatial derivative of ρ) to be well-conditioned, a requirement that is not met when the signal-to-noise ratio (SNR) of individual frames is low. The ensemble-mean displacement field, which has an effective SNR approximately $\sqrt{N}$ times that of a single frame, is generally amenable to Poisson inversion.

2.5 Natural Convection as a Validation Case

Natural convection from a heated vertical or horizontal plate is a classical, well-characterized flow for which both analytical similarity solutions and empirical correlations (e.g., Churchill and Chu) exist. Hargather and Settles [19] applied BOS to measure density gradients in natural convection and compared with theoretical predictions, demonstrating sub-percent-level agreement with the expected boundary-layer thickness and temperature profiles. This makes natural convection from a heated plate an ideal validation case for an RSD reconstruction pipeline: the expected density-gradient magnitude, boundary-layer thickness, and temperature profile shape are all known independently.

2.6 Research Gaps and Motivation

The following gaps motivate the present work and are directly addressed by the contributions described in subsequent chapters:

i. Quantitative RSD data for a compressible wall jet over a heated flat plate is essentially absent from the literature.

ii. No documented, open-source software library exists for the complete Greenberg RSD pipeline (calibration, processing, reconstruction), *pyrsd* fills this gap.

Chapter 3: METHODOLOGY

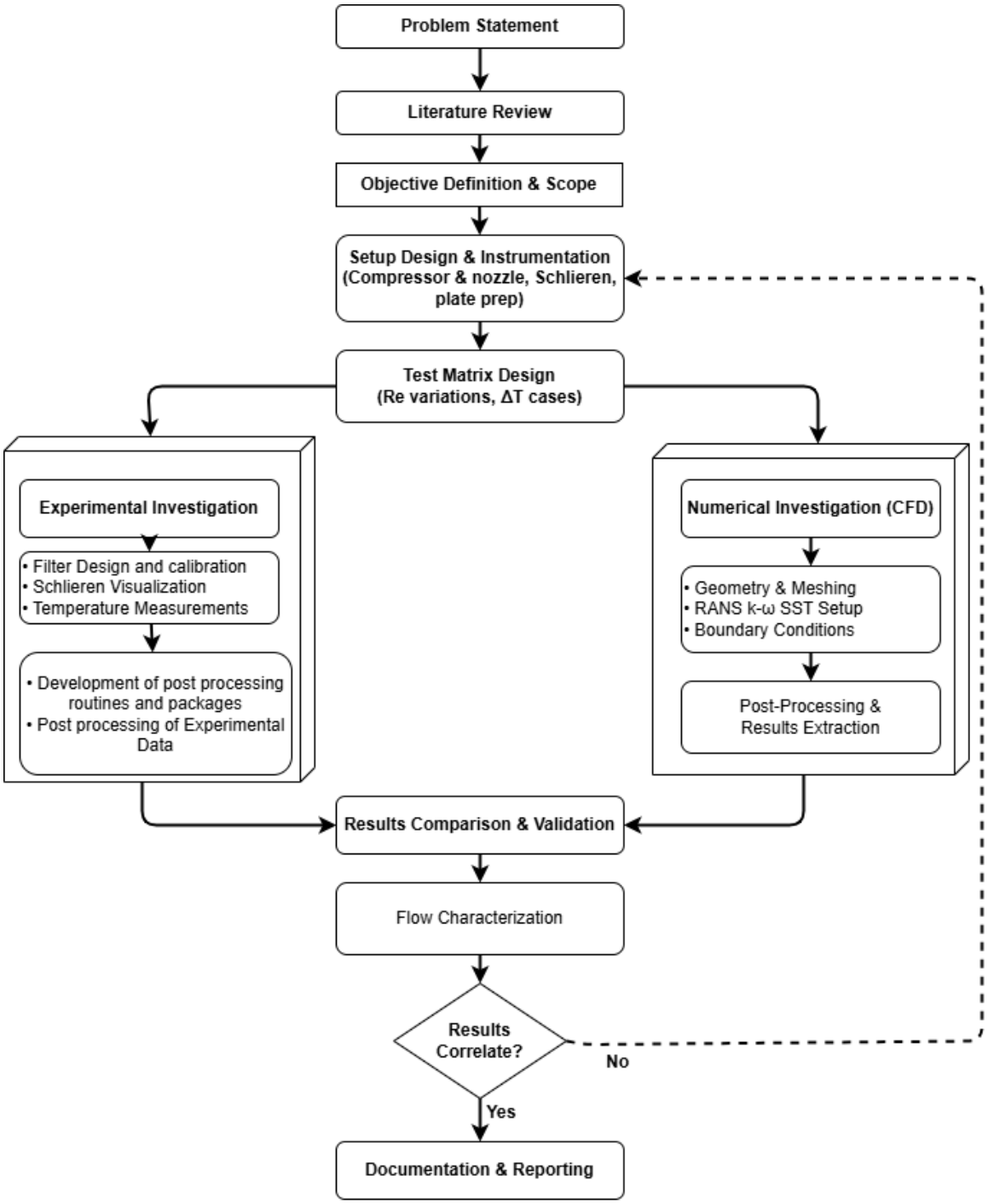


Figure 3.1: Overall methodology of the present investigation.

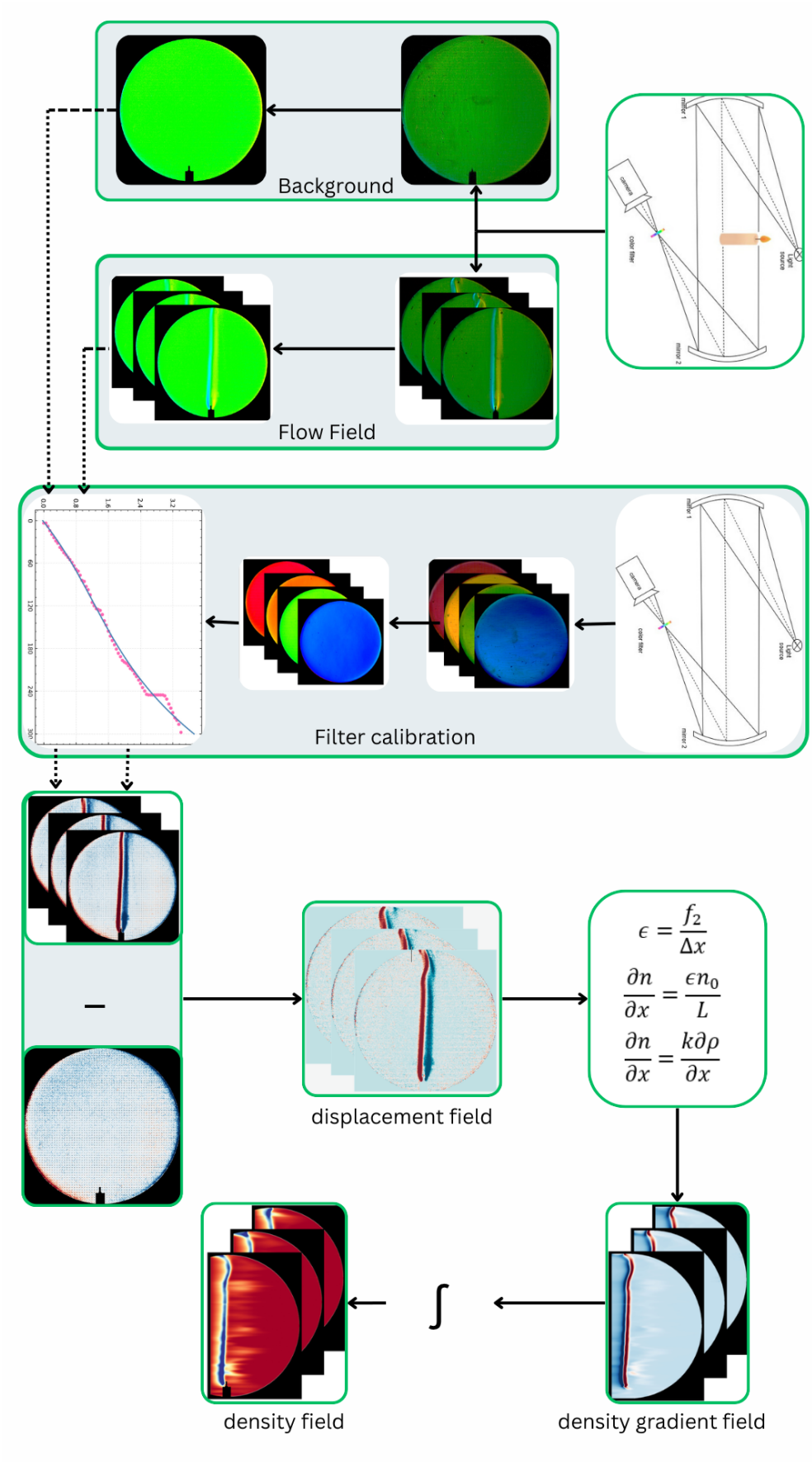


Figure 3.2: experimental and post processing work flow.

3.1 Experimental Setup

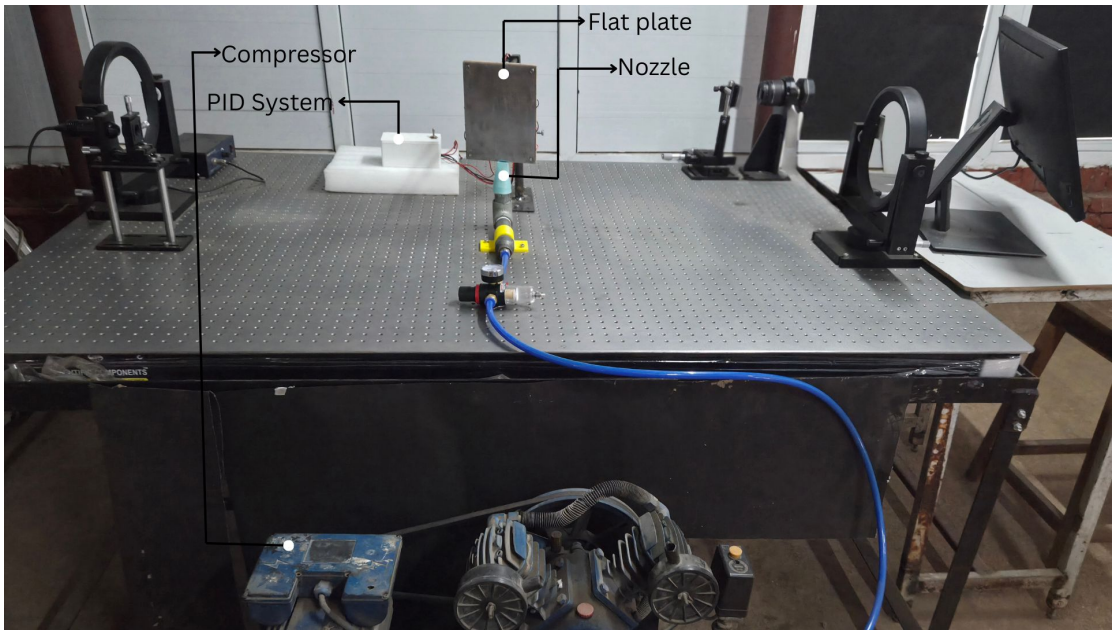


Figure 3.3: Experimental facility. The plate and nozzle assembly.

The experimental facility generates a controlled high-subsonic compressible jet which interacts tangentially with a temperature-controlled flat plate. Due to the physical layout of the Z-type schlieren system and constraint of space on schlieren system available, a horizontal test section would have obstructed the direct optical path. So, the plate and nozzle assembly is oriented vertically with the jet issuing upward. All results are presented in the conventional horizontal orientation (jet flowing left to right).

3.1.1 Compressed Air Supply and Pressure Estimation

A 2 HP, 70-liter reservoir compressor supplies compressed air. The supply path from the reservoir to the nozzle inlet consists, in series, of approximately 1 m of flexible hose (8 mm internal diameter), a Bourdon-tube pressure regulator with analogue gauge, approximately 15 cm of flexible hose (8 mm ID), a 10 cm section of 1 inch cast-iron pipe, a 90° 1 inch elbow turning the flow vertically, and the 3D printed converging nozzle. The Bourdon gauge measures *gauge* pressure at the regulator outlet relative to the local ambient; it does not measure the total pressure at the nozzle inlet.

Significant total pressure losses arise along this path due to pipe friction in the 8 mm hose, the 90° elbow (loss coefficient $K \approx 0.9$), and area change transitions. CFD simulation of

the full supply plumbing (Section 3.5) at 1.0 bar gauge yielded a nozzle exit velocity of approximately 260 m/s, implying a nozzle inlet total pressure $p_{0,\text{nozzle}} \approx 1.31$ bar absolute, a 30% reduction from the regulator absolute pressure of 1.87 bar. This loss factor is used to estimate the nozzle operating point at each setting.

At Kathmandu altitude (≈1300 m), the ambient static pressure $p_\infty \approx 0.871$ bar is significantly lower than the sea-level standard. For a converging nozzle, the throat chokes when the nozzle pressure ratio exceeds the critical value:

$$\text{NPR}_{\text{crit}} = \left(\frac{\gamma+1}{2}\right)^{\gamma/(\gamma-1)} = 1.893 \quad (\gamma = 1.4), \tag{3.1}$$

requiring a minimum nozzle inlet total pressure of $1.893 \times 0.871 = 1.649$ bar absolute. Accounting for approximately 30-40% supply-line loss, the nozzle chokes when the regulator gauge pressure exceeds approximately 1.9 bar. Table 3.1 summarizes the estimated exit conditions.

Table 3.1: Estimated nozzle exit conditions ($T_\infty = 289$ K, $p_\infty = 0.871$ bar, supply-line loss factor ≈ 0.70 from CFD of plumbing).

p_{gauge} (bar)	$p_{0,\text{regulator}}$ (bar abs)	$p_{0,\text{nozzle}}$ (bar abs)	NPR	M_{exit}	V_{exit} (m/s)
1.0	1.871	1.31	1.51	≈ 0.75	≈ 260
1.4	2.271	1.47	1.68	≈ 0.90	≈ 305

At 2 bar gauge and above, two sharp lines converging toward the jet centerline were visible in schlieren images, emanating from the corners of the rectangular nozzle lip. These structures intensify progressively from 2 bar through 4 bar gauge and are characteristic of oblique compression waves from an under-expanded supersonic jet, confirming that the nozzle is choked at these settings. No such structures are observed at 1.0 or 1.4 bar gauge. Quantitative RSD measurements were performed only at 1.0 and 1.4 bar reservoir gauge pressure.

3.1.2 Nozzle Design

A converging nozzle accelerates the flow from the 1 inch pipe ($A_{\text{inlet}} \approx 507$ mm^2) to a 12 mm × 5 mm rectangular exit ($A_{\text{exit}} = 60$ mm^2), giving a contraction ratio of 9.25. The

height profile follows a 5th-order polynomial, ensuring zero slope and curvature at both ends [20, 21]:

$$H(z) = H_1 - (H_1 - H_2)\left[6\left(\tfrac{z}{L}\right)^5 - 15\left(\tfrac{z}{L}\right)^4 + 10\left(\tfrac{z}{L}\right)^3\right], \tag{3.2}$$

with a linear width contraction $W(z) = W_1 - (W_1 - W_2)z/L$. The full quasi-one-dimensional isentropic design analysis is given in Appendix A. The nozzle was fabricated by fused deposition additive manufacturing.

Figure 3.4: 3D-printed converging nozzle (exit 12 mm × 5 mm, contraction ratio 9.25).

3.1.3 Heated Plate Assembly

The flat plate is fabricated from a high thermal conductivity metal to promote lateral heat spreading. Thermoelectric (Peltier) modules are mounted on the underside of the plate to provide heating. A layer of thermal paste is applied at the interface to minimize thermal contact resistance and ensure efficient conduction of heat from the modules to the plate; the cold side of each module is thermally coupled to an insulated part(plywood) to prevent cold-side cooling from influencing the flow. An Arduino Nano microcontroller running a PID algorithm regulates electrical power to the Peltier modules via a MOSFET,

Table 3.2: Comprehensive System Wiring Connections

Origin	Intermediate Component	Destination
A. The PID Driver		
Arduino D3	1 kΩ Resistor	Base of Q1
Q1 Emitter	-	GND
Q1 Collector	4.7 kΩ Resistor	+12V
Q1 Collector	-	Base of Q2
Q2 Emitter	-	GND
Q2 Collector	1 kΩ Resistor	+12V
Q2 Collector	-	4× 330 Ω Gate
B. The Power Stage (4× IRFZ44N MOSFETs)		
4× 330 Ω Gate Resistors	-	Gate of each MOSFET
All MOSFET Sources	-	ATX GND
Each MOSFET Drain	-	Peltier Negative (-)
C. The Heaters (4× TEC1-12706)		
Peltier Positive (+)	-	ATX +12V
Peltier Negative (-)	-	Respective MOSFET Drain
D. The Sensor (10k NTC Thermistor)		
+5V (Arduino)	-	Thermistor Leg 1
Thermistor Leg 2	-	Arduino A0
Arduino A0	9.5 kΩ Resistor	GND

stabilizing the surface temperature at setpoints of 40, 60, and 80°C. Embedded NTC thermistors near the plate surface provide continuous temperature feedback and log data to the serial output for confirmation of steady state conditions.

Infrared thermography was attempted to assess temperature uniformity. However, the polished metallic surface has very low thermal emissivity, causing the IR camera to image reflected ambient radiation rather than the plate surface. Consequently, IRT was not used for temperature validation.

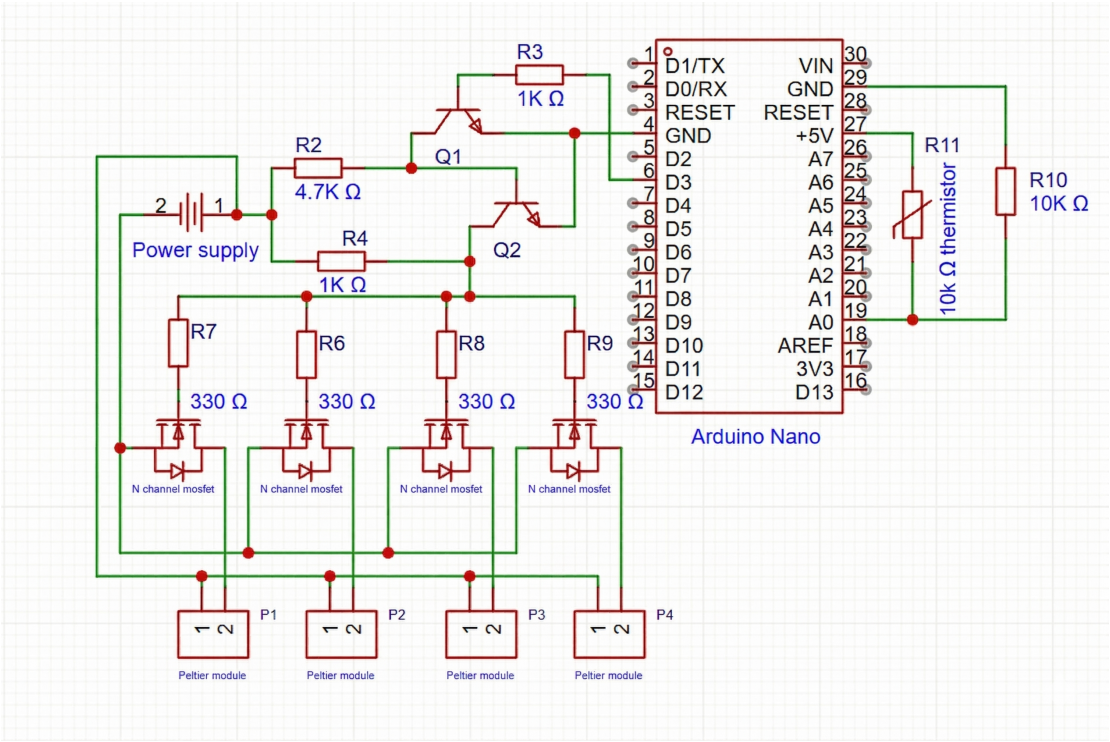


Figure 3.5: Schematic diagram of PID system

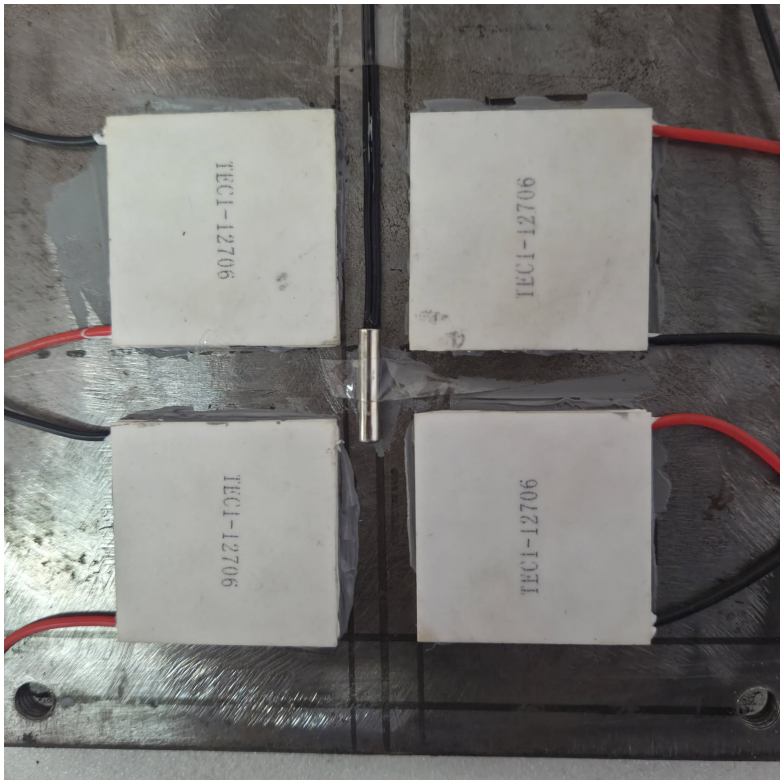


Figure 3.6: Peltier Module and thermistor assembly in the metal place.

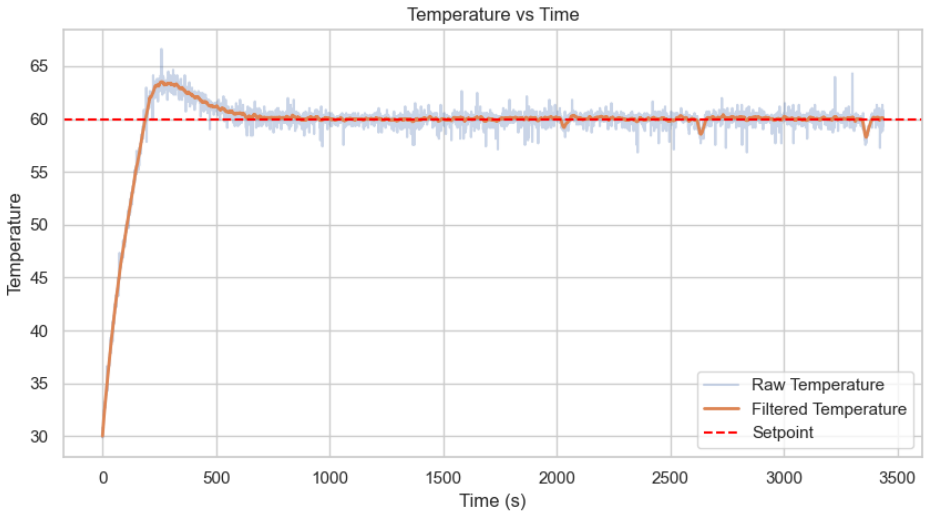


Figure 3.7: Response of PID system for 60°C

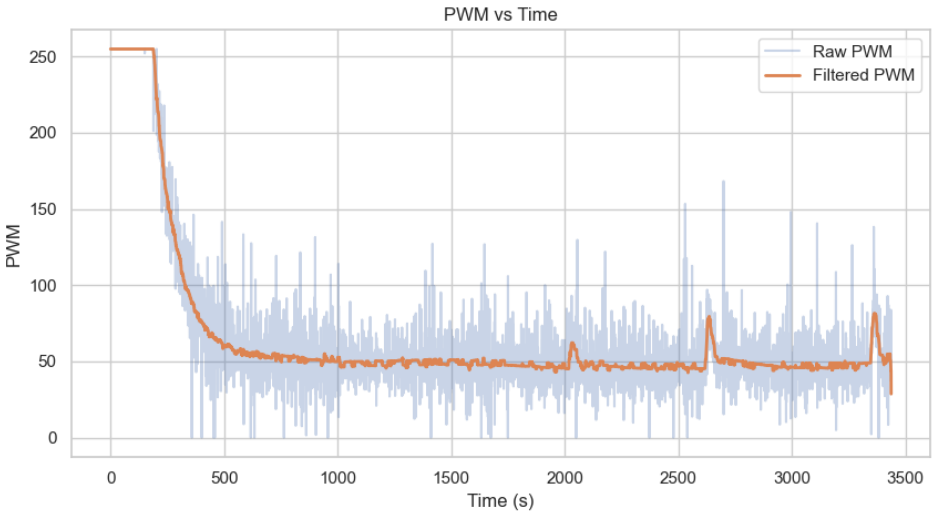


Figure 3.8: PWM response to the temperature.

3.2 Schlieren System

A double mirror Z-type schlieren system is used throughout this work, employing two parabolic mirrors of focal length $f = 1500$ mm and aperture 200 mm. The single mirror arrangement was evaluated and discarded because it created the issue of double images. The knife edge could be used effectively only in the vertical cutoff orientation. Horizontal cutoff was not achievable with the available hardware.

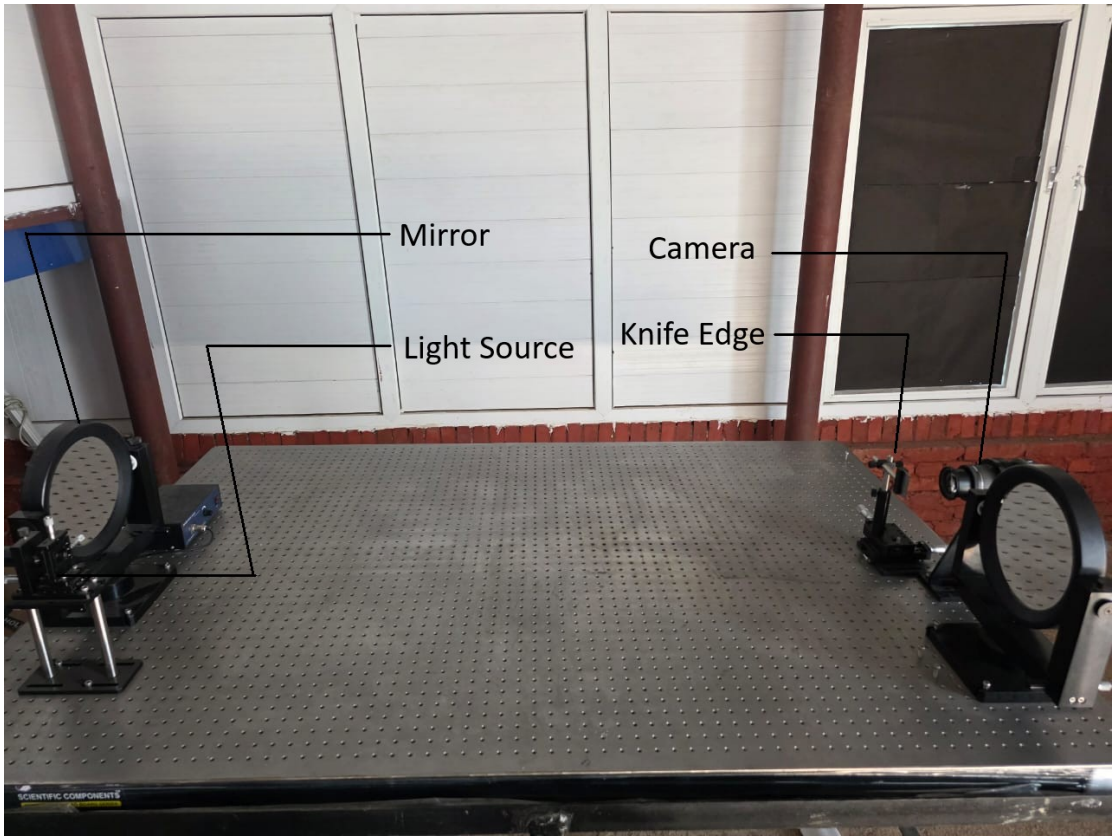


Figure 3.9: Complete experimental setup with Z-type double-mirror schlieren system.

3.2.1 Aperture design and fabrication

Traditional schlieren systems use a point or circular pinhole source to visualize optical inhomogeneities in a flow field, but this only yields a total deflection angle that cannot be directly related to scalar flow properties such as refractive index or density. A rectangular source aperture addresses this limitation by allowing either the transverse or axial deflection angle to be isolated depending on the aperture orientation, enabling a quantitative relationship to the refractive index field through Gladstone-Dale relation. [5]. So, for this reason a rectangular source aperture was design and fabricated using 3D

printing method. For the edge of the aperture, sharp edged blade was used. the width of the aperture was estimated about 0.15 mm using spatially calibrated image.

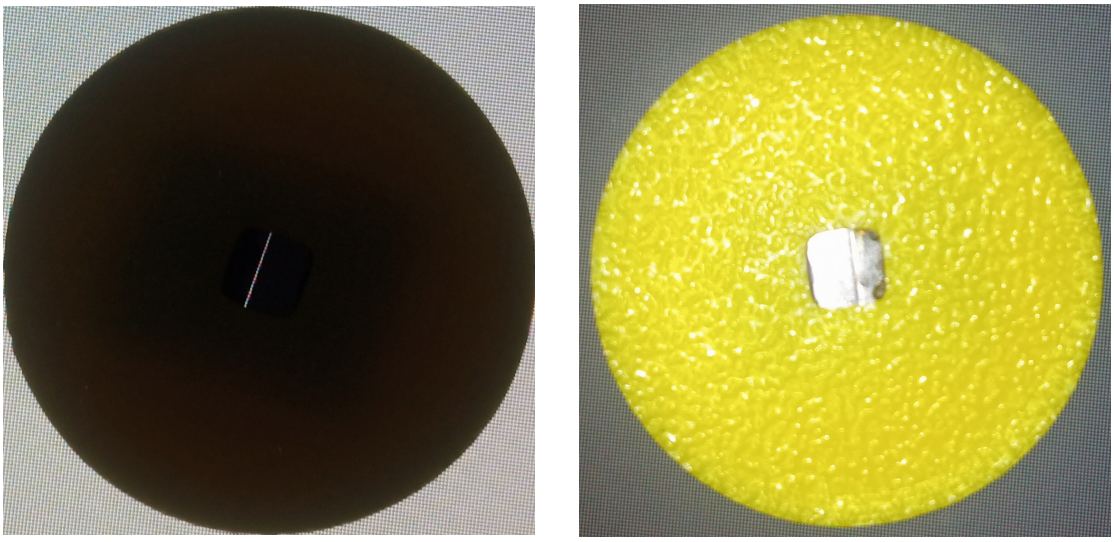


Figure 3.10: Rectangular source aperture used in RSD

3.2.2 Filter Design, Generation, and Fabrication

The color filter placed at the focal plane of the schlieren system is the central component of RSD. It converts a spatial displacement of the focused ray bundle into measurable hue change. Basically, it functions as a very fine ruler that can resolve very tiny displacements of light in focal plane. The filter can be designed such that it can measure deflection along one dimension, both dimension, or magnitude of deflection. The filter that can measure deflection in both direction requires xz translation stage. Due to this constraint, filter that can measure only one dimension was designed. [2] have well established that the only single parameter should vary along the axis of measurement in the filter and it should be independent of properties of light and other system parameters for optimum measurements. This is best achieved through HSI based systems [2]. HSI based systems are particularly use in contrast to RGB based systems as they decouple the measurement from any intensity variations due to uneven lighting or light attenuation [2]. To design the color filter, the polar angles of hue values are mapped onto one dimensional coordinate for measuring deflection along one dimension and are mapped onto a radial coordinate, an axisymmetric, bullseye filter is produced that measures magnitude of deflection [2]. The filter design procedure involves selection of hue range, width of the filter, and the fabrication medium for the filter which determines sensitivity and quality of the results.

Hue range. The HSV hue axis spans 0° through 360°, but hues near 360° wrap back

to red, which is identical in appearance to hues near 0°. This ambiguity makes the calibration spline non-invertible if hue wraps within the valid measurement range. That is, it becomes challenging to distinguish if light deflected in positive or negative direction for that hue to appear. This is also the primary reason for not to choose symmetric filter for asymmetric flows. Due to this reason, hue range of 0° to 300° (red through magenta, excluding the red wrap-around zone) was chosen. A secondary series using 0° to 360° was included for comparative evaluation, but was not used in quantitative measurements due to the aforementioned ambiguity.

Filter width and sensitivity. The sensitivity of the RSD system, the hue change per unit ray deflection, is inversely proportional to the filter width. A narrower filter produces a larger hue change per unit displacement at the focal plane, but saturates (reaches the end of the color range) at a smaller maximum deflection. The maximum measurable deflection $\varepsilon_{\max}$ is related to the filter half-width $w/2$ and the second mirror focal length f_2 by $\varepsilon_{\max} = w/(2f_2)$ [2]. For the present system with $f_2 = 1500$ mm, a 3.0 mm filter gives $\varepsilon_{\max} \approx 1.0$ mrad and a 1.5 mm filter gives $\varepsilon_{\max} \approx 0.5$ mrad. The wall-jet density gradients at 1.0 bar are moderate, favoring a mid-range width. The widths included in the fabricated filters step from 1.5 mm to 5 mm in 0.5 mm increments, allowing systematic characterization.

Saturation. A saturation of 1.0 (fully saturated) gives the highest color purity and the sharpest hue transitions, but printed colors at saturation 1.0 may not turn out as expected depending upon the printing services. So, few series of filters at saturation 0.5 were therefore included in Row 2 of Slide 1 for comparison; all quantitative measurements used the saturation 1.0 series, which produced cleaner calibration curves.

Fabrication medium. Initial filters were digitally printed on transparency film using a color laser printer. These proved optically inadequate. The printed color gradients exhibited visible halftone dot structure at the scale of the focal spot ($\sim$0.5 mm), introducing periodic hue noise into the calibration and very poor transmissivity. The filters were subsequently printed on E6 reversal (slide) film at a professional photo lab through digital image projection services. E6 slide film has a grain size of approximately 3 μm to 5 μm, more than two orders of magnitude finer than the focal spot, and the dye-based color process reproduces continuous gradients without halftone artifacts. The resulting filters produced smooth, monotonic calibration curves with no visible banding and superior transmissivity.

Slide layout. Three filter slides were designed and printed on E6 film. (Fig. 3.11). **Slide 1** contains continuous one-dimensional color gradient filters (hue 0-300°, saturation 1.0)

of varying widths. Row 1 spans widths from 5.0 mm to 1.5 mm in 0.5 mm steps. Row 2 repeats the width series with saturation 0.5 for six filters and hue range 0-360° with 1.0 mm width increments for remaining three filters. **Slide 2** contains color filters overlaid with symmetric grey scale gradients for improved weak and small gradient capturing. The circular bull's eye filters (deflection magnitude only) measure deflection magnitude without measuring direction. Both of these filter can be calibrated filter transversal by primarily measuring hue for quantitative measurements. **Slide 3** contains discrete color strips and hue saturation reference targets for calibration verification and color calibration for raw images. While designing filters, the hues were distributed along the width w of the filter within desired hue range. The height of filters do not affect anything if its height is higher than that of source slit. A python program was used to create each filter which was laid out in 34 by 24 aspect ratio high resolution tiff image. The images were developed through the service of professional slide film studio, digitalslides.co.uk, based on UK.

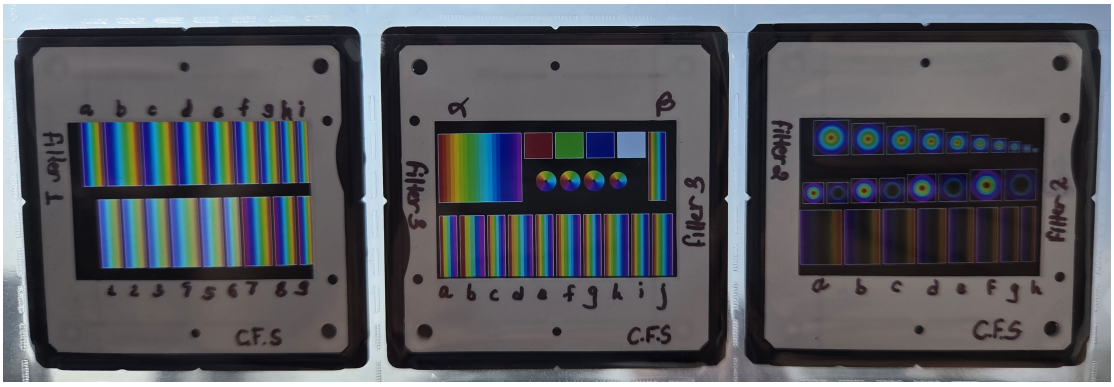


Figure 3.11: The three slides with filter of varying widths and hue ranges

3.2.3 Filter Selection for Quantitative Measurements

Filters with width ranging from 1.5 mm to 3 mm from Slide 1 with the rectangular slit source aperture were used for all quantitative RSD measurements. Narrower filters provide higher sensitivity (larger hue change per unit deflection) but saturate more readily in high-gradient regions near the wall. Various source geometries were evaluated to optimize sensitivity. A rectangular aperture yielded noticeably higher schlieren contrast than a circular pinhole of equivalent area, confirming the directional sensitivity advantage reported by Agrawal *et al.* [5] and Hargather *et al.* [19].

3.2.4 Conventional (Knife-Edge) Schlieren

Knife-edge schlieren was used for qualitative visualization and system validation. The vertical cutoff makes the system sensitive to horizontal density gradients ($\partial\rho/\partial x$). Cold wall jet cases produce faint structures; heated wall cases at 60 and 80°C produce clearly visible density-gradient signatures. Sensitivity increases with wall temperature consistent with the proportionally larger density change at higher wall superheat.

3.2.5 Rainbow Schlieren Deflectometry - Calibration

A filter is mounted on the micrometer translation stage at the focal plane. Following Greenberg *et al.* [2], the filter is displaced in known increments with no flow. The images are cleaned by cropping, masking and aligning them. A small window size at the center of the image is defined where *pyrsd.build_calibration_data*[22] records the circular mean hue in a central ROI at each step, producing a hue to displacement table. A smoothing spline is fitted to the valid hue range 0-300°. Data above 300° are excluded because the filter saturates and hue wraps around (Fig. 3.12).

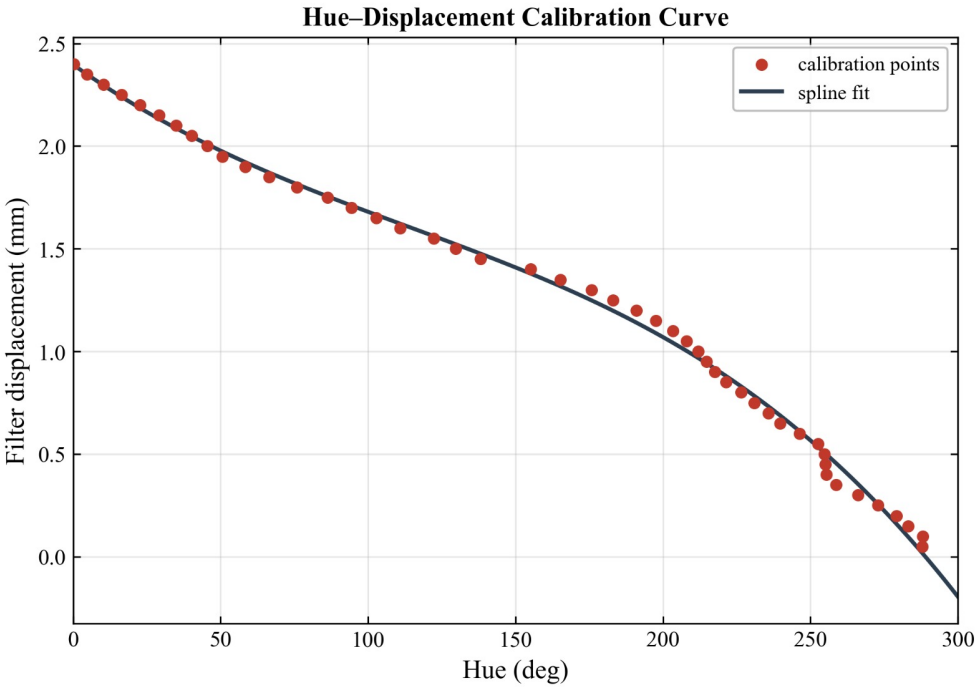


Figure 3.12: RSD calibration curve: hue versus filter displacement.

3.2.6 RSD Measurement Principles

Refractive Index and the Gladstone-Dale Relation

The operating principle of Rainbow Schlieren Deflectometry is based on the relationship between the refractive index of a gas and its local density. For a single component gas, this is given by the Gladstone Dale relation.

$$n - 1 = K_{\mathrm{GD}}\,\rho \tag{3.3}$$

where n is the refractive index, ρ is the local density ($\mathrm{kg\,m^{-3}}$), and K_{GD} is the Gladstone Dale constant. For air at visible wavelengths, $K_{\mathrm{GD}} = 2.23 \times 10^{-4}\ \mathrm{m^3\,kg^{-1}}$ [16]. Differentiating with respect to the wall normal coordinate y gives the working form used in all subsequent reconstructions.

$$\frac{\partial n}{\partial y} = K_{\mathrm{GD}}\,\frac{\partial \rho}{\partial y} \tag{3.4}$$

Ray Deflection in a Non Uniform Medium

A collimated ray propagating along the optical axis z (the spanwise direction in the present experiment) through a medium characterized by a transverse refractive index gradient undergoes cumulative angular deflection. For a medium of uniform spanwise depth L, the total deflection angle ε in the wall normal direction is obtained by integrating the ray equation along the path [2, 23].

$$\varepsilon = \frac{1}{n_0}\int_0^L \frac{\partial n}{\partial y}\,\mathrm{d}z \approx \frac{L}{n_0}\,\frac{\partial n}{\partial y} \tag{3.5}$$

where $n_0 \approx 1.000293$ is the ambient refractive index of air and the line of sight uniformity approximation holds when the flow is quasi 2D over the spanwise depth $L = 150\,\mathrm{mm}$ captured by the schlieren field of view. Substituting Eq. (3.4) into Eq. (3.5).

$$\varepsilon = \frac{K_{\mathrm{GD}}\,L}{n_0}\,\frac{\partial \rho}{\partial y} \tag{3.6}$$

This is the fundamental measurement equation. The angular deflection is directly proportional to the spanwise averaged wall normal density gradient. The line of sight integration introduces an inherent spatial averaging that must be considered when comparing RSD results with point wise CFD predictions.

Filter Displacement and Hue Encoding

In the Z type double mirror schlieren system, the deflected ray bundle converges at the focal plane of the second parabolic mirror, where the rainbow filter is mounted in a linear translation stage. A ray deflected by angle ε at the test section arrives at the filter displaced from its undeflected position by:

$$\Delta d = f_2 \, \varepsilon \tag{3.7}$$

where $f_2 = 1500\,\text{mm}$ is the focal length of the second mirror in current setup. The continuously varying hue of the filter encodes this displacement as a measurable color change. The calibration procedure (Section 3.2.6) establishes the invertible lookup function $\mathscr{H}(\Delta d)$ mapping displacement to hue, so that for any image pixel at location (x,y):

$$\Delta d(x,y) = \mathscr{H}^{-1}\big[h_{\text{flow}}(x,y)\big] - \mathscr{H}^{-1}\big[h_{\text{bg}}(x,y)\big] \tag{3.8}$$

where h_{flow} and h_{bg} are the hue fields recorded with and without flow respectively. The differencing step removes spatial non uniformness in the filter arising from manufacturing imperfections and non uniform illumination.

Density Gradient Computation

Combining Eqs. (3.6) and (3.7), the wall normal density gradient at each pixel is:

$$\frac{\partial \rho}{\partial y}(x,y) = \frac{n_0}{K_{\text{GD}} L f_2} \cdot \Delta d(x,y) \tag{3.9}$$

where n_0, K_{GD}, L, and f_2 depend only on fixed optical and geometric parameters for the experiment. For the present experimental study:

$$C = \frac{n_0}{K_{\text{GD}} L f_2} = \frac{1.000293}{2.23 \times 10^{-4} \times 150 \times 1500} = 0.0199 \text{ kg m}^{-3}\text{mm}^{-2} \tag{3.10}$$

Density Field Reconstruction by One Dimensional Integration

With the measured gradient field $\partial\rho/\partial y$, the density field can be recovered by cumulative trapezoidal integration in the wall normal direction, anchored at a reference column in the undisturbed ambient region where the density is known from the ideal gas law. This,

of course, assumes density gradient in along orthogonal direction is insignificant. For flows like natural convection over heated plate or rise of buoyant plumes, it holds more strongly. In most of the other cases, it can be very unrealistic. In current primary study of jet wall, it has been used cautiously.

$$\rho_\infty = \frac{p_\infty}{R_{\text{specific}}\, T_\infty} \tag{3.11}$$

At $p_\infty = 0.871\,\text{bar}$ and $T_\infty = 289\,\text{K}$, $\rho_\infty = 1.050\,\text{kg}\,\text{m}^{-3}$. The density at wall normal position y from the reference location y_{ref} is:

$$\rho(x,y) = \rho(x,y_{\text{ref}}) + \int_{y_{\text{ref}}}^{y} \frac{\partial \rho}{\partial y'}\,\mathrm{d}y' \tag{3.12}$$

In practice, this integral is evaluated by the trapezoidal rule applied column by column. The primary limitation of this approach is that integration error accumulates monotonically along each column. Noise in the gradient field produces characteristic streaks in the reconstructed density.

Density Field Reconstruction by Two-Dimensional Poisson Inversion

A globally more accurate reconstruction can be obtained by treating the problem as a Poisson boundary value problem [7] if both gradient component are available. Given both gradient components $\partial\rho/\partial x$ and $\partial\rho/\partial y$, the density field satisfies:

$$\nabla^2\rho = \frac{\partial^2\rho}{\partial x^2} + \frac{\partial^2\rho}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial\rho}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\rho}{\partial y}\right) = \nabla\cdot(\nabla\rho) \tag{3.13}$$

where the right hand side is assembled from the measured gradient fields. A single Dirichlet boundary condition $\rho = \rho_\infty$ is imposed at the ambient reference pixel. All remaining boundary pixels are treated with homogeneous Neumann conditions. The resulting sparse linear system is solved directly using the `scipy.sparse.linalg` framework within `pyrsd`. Because both gradient components contribute simultaneously, the Poisson inversion distributes reconstruction error globally rather than accumulating it. A practical requirement of the Poisson approach is that the right hand side of Eq. (3.13), the divergence of the gradient field, equivalent to the second spatial derivative of ρ, must be well-conditioned.

3.3 pyrsd: RSD Processing Library

pyrsd was developed as part of this project to provide a documented, reproducible processing pipeline for RSD data. No equivalent open-source library for post processing RSD results were available during the project interval. The library is available at `https://github.com/FVL2082/pyrsd` [22]. Table 3.3 summarizes the principal functions.

Table 3.3: *pyrsd* principal functions.

Function	**Description**
`build_calibration_data`	Loads sequential filter step images; extracts circular-mean hue (handles $0^\circ/360^\circ$ wrap around) from a central ROI; returns hue-displacement table.
`fit_spline`	Fits a smoothing `UnivariateSpline` to (hue, displacement) pairs within a specified valid hue range; saves calibration JSON.
`process_stack`	Loads flow image sequence and reference background; computes per-frame Δd; returns 3-D stack ($N \times H \times W$).
`ensemble_mean` / `std` / `turbulence_intensity`	Statistical calculations of the displacement stack.
`density_from_gradient_1d`	1-D cumulative integration (trapezoidal rule) anchored at a reference ambient pixel.
`density_from_gradient_2d`	2-D Poisson inversion via sparse direct solver or iterative solver, one Dirichlet anchor.
`temperature_ideal_gas` / `isobaric`	Converts density to temperature.
`gaussian_filter` / `median_filter` / `bilateral_filter`	filtering of displacement fields before reconstruction.

Design Principles Following design principles guided the development of `pyrsd`:

i. **Reproducibility.** Every calibration run produces a JSON file saving the filter name, step size, hue range, spline knots, and all processing parameters. Post processing scripts can load this file directly, ensuring that any result can be reproduced exactly from the raw images.

ii. **NaN awareness** Pixels outside the valid hue range (e.g., saturated near-wall pixels where the filter wraps beyond 300°, or pixels masked by the plate surface) are assigned NaN throughout. All arithmetic operations, ensemble statistics, and the solver propagate NaN masks without contaminating adjacent valid pixels.

iii. **Modularity.** Each processing step is a standalone function with a defined input and output, allowing individual steps to be replaced or extended without modifying the rest of the code. For example, the solver backend can be switched between a sparse direct solver and a conjugate gradient iterative solver by changing a single argument.

3.4 RSD Processing Pipeline

Step 1 - Calibration. Filter displaced in known steps with no flow; *pyrsd.build_calibration_data* records mean hue at each step; *pyrsd.fit_spline* fits the valid hue range.

Step 2 - Displacement extraction. Each flow frame is converted to HSV; the calibration spline is applied to both the flow hue and a no-flow background, and their difference gives $\Delta d(x,y)$ for that frame. Repeated for all N frames.

Step 3 - Ensemble averaging. *pyrsd.ensemble_mean* computes $\overline{\Delta d}(x,y)$ over all N frames.

Step 4 - Density-gradient computation. The mean displacement is converted to a density-gradient field using the setup constant

$$C = \frac{n_0}{f_2 \cdot L \cdot K_{\mathrm{GD}}}, \tag{3.14}$$

where $n_0 \approx 1.000293$ is the ambient refractive index, $f_2 = 1500$ mm, $L = 150$ mm is the spanwise test-section depth, and $K_{\mathrm{GD}} = 2.23 \times 10^{-4}$ m^3 kg^{-1} for air [16]. The line of sight integrated density gradient is $\partial\rho/\partial y = C \cdot \overline{\Delta d}$.

Step 5 - Density reconstruction. column by column cumulative 1D integration along the wall-normal direction is performed, anchored at the ambient column where $\rho_\infty = p_\infty/(RT_\infty)$.

Step 6 - Temperature. Under the isobaric approximation (appropriate for the buoyancy-driven natural convection case and the far from stagnation wall jet region): $T = T_\infty \rho_\infty/\rho$.

3.4.1 Ensemble Convergence Verification

Early frames in each acquisition are excluded before ensemble averaging because flow startup can bias the mean field. A rolling coefficient of variation (RCoV) is computed from the per frame domain mean absolute displacement $\bar{d}_i = \langle|\Delta d|\rangle_i$ over a sliding window of $W = 50$ frames:

$$\mathrm{RCoV}(i) = \frac{\mathrm{std}\big(\bar{d}_{i-W/2}, \ldots, \bar{d}_{i+W/2}\big)}{\mathrm{mean}\big(\bar{d}_{i-W/2}, \ldots, \bar{d}_{i+W/2}\big)} \tag{3.15}$$

A low RCoV indicates that the per frame displacement signal is stationary and the mean is no longer evolving. Frames are included in the ensemble only after the RCoV has dropped and remained below 5% for at least 50 consecutive frames. The mean is then computed over all frames in the stable window.

3.5 CFD Methodology

Governing Equations

The flow is governed by the compressible Reynolds Averaged Navier-Stokes (RANS) equations expressing conservation of mass, momentum, and energy. Written in conservative form for steady flow:

Continuity

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \tag{3.16}$$

Momentum

$$\frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu_{\text{eff}}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial u_k}{\partial x_k}\right)\right] \tag{3.17}$$

where $\mu_{\text{eff}} = \mu_{\text{lam}} + \mu_t$ is the sum of the laminar and turbulent (eddy) dynamic viscosities.

Energy

$$\frac{\partial(\rho u_i H)}{\partial x_i} = \frac{\partial}{\partial x_i}\left[\left(\frac{\lambda}{c_p} + \frac{\mu_t}{\text{Pr}_t}\right)\frac{\partial H}{\partial x_i}\right] \tag{3.18}$$

where $H = h + \frac{1}{2}u_i u_i + k$ is the specific total enthalpy, λ is the thermal conductivity, c_p is the specific heat at constant pressure, and the turbulent Prandtl number $\text{Pr}_t = 0.9$ is used throughout following standard practice for wall bounded compressible flows.

Equation of state Air is treated as an ideal gas:

$$\rho = \frac{p}{R_{\text{specific}}\,T}, \qquad R_{\text{specific}} = 287\ \text{J}\,\text{kg}^{-1}\,\text{K}^{-1} \tag{3.19}$$

3.5.1 Mesh Independence Test

A mesh independence test was conducted to ensure that the numerical results are not influenced by the density of the computational mesh. In finite-element and finite-volume solvers such as ANSYS Fluent, the discretizations of the flow domain introduces a truncation error that decreases as the mesh is refined. However, beyond a certain refinement level, further reduction in element size yields negligible change in the solution while increasing computational cost significantly.

Plumbing

In the present study, the plumbing assembly was meshed at six different global element sizes ranging from 8.0 mm to 1.0 mm. The nozzle outlet velocity was selected as the monitoring parameter, as it is the primary quantity of interest and is sensitive to mesh resolution near the nozzle exit. All other boundary conditions and solver settings were held constant across all mesh configurations.

Table 3.4: Mesh independence test results for the plumbing assembly

Element Size (mm)	**No. of Elements**	**Nozzle Outlet Velocity (m/s)**	**Change from Previous (%)**
8.0	97278	256.1818	—
5.0	116288	257.8299	0.6434
3.0	153003	258.9033	0.4164
2.0	278710	260.3153	0.5453
1.5	517104	260.4467	0.0505
1.0	1464343	260.4928	0.0177

As shown in Table 3.4, the nozzle outlet velocity converges as the element size is reduced. The solution varies by less than 1 % between Mesh all mesh level, indicating that all Mesh Level provides a mesh-independent solution. Accordingly, an element size of 2 mm was adopted for all subsequent simulations, balancing solution accuracy with computational efficiency.

Wall-Jet

For the wall-jet configuration, the 2-D domain was meshed at four different refinement levels. The number of divisions along each edge was increased by a factor of $\sqrt{2}$ at each level, effectively doubling the total element count from one mesh to the next. The Wall Heat Transfer Coefficient (HTC) was selected as the convergence parameter, as it is directly governed by the near-wall temperature gradient and is highly sensitive to mesh resolution in the region of interest. All boundary conditions and solver settings were kept identical across all four mesh configurations.

Table 3.5: Mesh independence test results for the wall-jet configuration.

No. of Elements	**HTC ($\mathrm{W\,m^{-2}\,K^{-1}}$)**	**Change from Previous (%)**
35,400	362.75	—
68,949	367.94	1.432
140,400	374.64	1.819
278,052	375.93	0.345

The results show that the HTC value changes by less than 0.35% between the 140,400 and 278,052 element meshes, indicating that the solution is effectively mesh-independent at this refinement level. The 140,400 element mesh was therefore selected for all subsequent simulations as it provides a good balance between computational cost and solution accuracy.

3.5.2 Plumbing Simulation

The full supply path from the pressure regulator to the nozzle exit was modelled as a steady-state compressible flow in ANSYS Fluent using a pressure-based coupled solver. Air was treated as an ideal gas (Eq. 3.19), which is necessary to correctly capture the density variation between the high-pressure supply line and the nozzle exit at $Ma \approx 0.75$. Since no external heating is applied to the pipe walls, all solid boundaries were set as adiabatic no-slip walls. Turbulence closure was provided by the realizable k-ε model with Standard Wall function. [24]. For internal ducted flows, such as the present supply plumbing comprising straight runs, bends, and a convergent nozzle, the realizable variant with standard wall function offers improved predictions of round-jet spreading and flows with strong streamline curvature compared to the standard k-ε formulation [24]. The purpose of this simulation is solely to determine the nozzle exit Mach number and total conditions for use as inlet boundary conditions in the wall-jet simulation.

The inlet was specified as a pressure inlet at a gauge pressure of 0.1 MPa (absolute 0.1871 MPa) with reference temperature of $T = 297$ K, representative of the compressed air supply at room temperature. The outlet was set to a gauge pressure of 0 Pa (absolute 87,100 Pa), consistent with the Kathmandu ambient condition. The operating pressure was set to 87,100 Pa throughout, matching the wall-jet simulation.

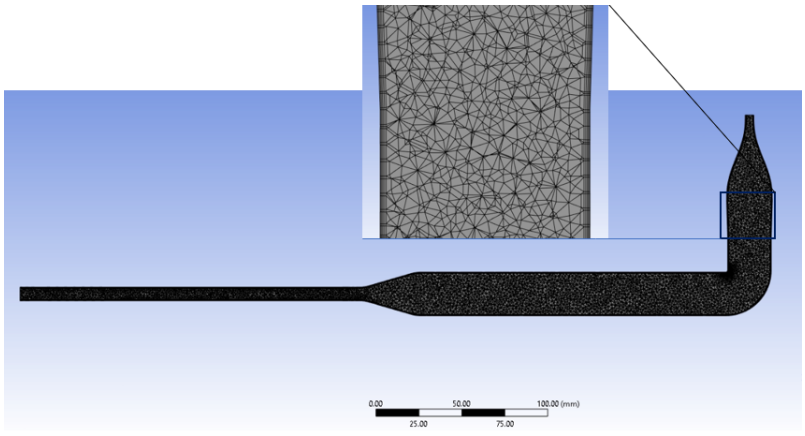


Figure 3.13: Meshing of plumbing assembly

The converged solution provides the nozzle exit velocity, Mach number, and total temperature, which are subsequently used as the inlet boundary conditions for the wall-jet simulation.

3.5.3 Wall-jet Simulation

A 2-D domain from the nozzle exit to the downstream region over the plate. The 2-D approximation is justified because the nozzle spanwise aspect ratio is 2.4, and RSD is itself a line-of-sight integral over $L = 150$ mm, averaging out moderate 3-D variations. The inlet condition is the Mach number and total temperature from the plumbing simulation; the outlet is the Kathmandu ambient pressure $p_\infty = 0.871$ bar. The plate upper surface is held at the experimental isothermal temperature. The density field from the converged solution is differentiated to obtain $|\nabla\rho|$ for direct comparison with the RSD density-gradient field.

The computational domain extends 1000 mm in the streamwise direction and 200 mm in the wall-normal direction. A fully structured mesh comprising 140 400 elements was employed. Near-wall mesh refinement was applied such that the dimensionless wall distance y^+ is maintained at approximately unity ($y^+ \approx 1$), ensuring that the viscous sub-layer is fully resolved without recourse to wall functions.

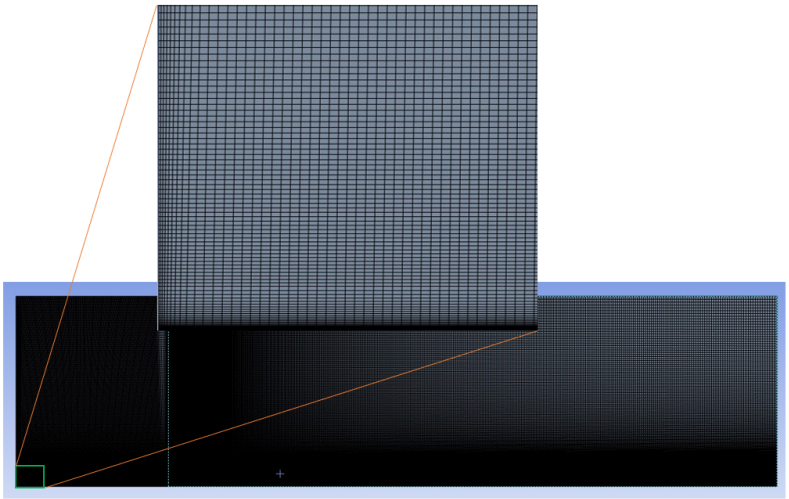


Figure 3.14: Meshing of 2-D domain

Solver Configuration

Steady-state compressible flow simulations were performed in ANSYS Fluent using a pressure-based coupled solver with the energy equation enabled. Air was modelled as an ideal gas, with density computed from the ideal gas law (Eq. 3.19). This treatment is necessary at the present operating Mach number of $Ma \approx 0.75$, where compressibility effects are non-negligible and the assumption of constant or incompressible density would introduce significant error in both the velocity and thermal fields. The operating pressure was set to $p_{\text{ref}} = 87{,}100$ Pa, corresponding to the ambient atmospheric pressure at the experimental site in Kathmandu ($\approx$ 1,300 m elevation). All gauge pressures in the simulation are referenced to this value.

Turbulence Model

Turbulence closure is modeled using the k-ω Shear Stress Transport (SST) formulation [25, 26]. This model blends near-wall k-ω accuracy, enabling direct integration at $y^+ \approx 1$ without damping functions with the freestream stability of k-ε [27]. Key features include a Bradshaw-based shear stress limiter to prevent eddy viscosity overproduction under adverse pressure gradients and a compressibility correction for $Ma \approx 0.77$ [28]. Comparative RANS studies confirm that SST provides superior agreement with experimental Stanton numbers for heated wall jets over standard k-ω or k-ε models [11, 29].

Boundary Conditions

Table 3.6: Boundary conditions for the 2-D wall-jet simulation

Boundary	**Type**	**Thermal Condition**	**Notes**
Nozzle outlet	Velocity inlet	$T = 297$ K (24 °C)	$U = 260$ m/s
Heated plate	No-slip wall	$T_w = 333$ K (60 °C)	Dirichlet condition
Far-field/Top	Pressure outlet	$T = 299$ K (26 °C)	Gauge $P = 0$ Pa

The heated plate is an isothermal no-slip wall ($T_w = 333$ K), matching experimental thermocouple data. Far-field and top boundaries are defined as pressure outlets at the Kathmandu ambient pressure (absolute $87,100$ Pa), allowing natural air entrainment into the domain without artificial co-flow constraints.

3.6 Test Matrix

Table 3.7: Wall jet test matrix.

p_{gauge} (bar)	T_w (°C)	Purpose
1.0	40	Wall jet; low heating
1.0	60	Wall jet; higher heating
1.4	40	Wall jet; higher NPR; low heating
1.4	60	Wall jet; higher NPR; high heating

Experiments were performed at 26°C ambient temperature with 24°C jet temperature at exit. All four runs used filter with 2.5 mm width and 0° to 300° hue range and rectangular slit source.

3.7 Sources of Uncertainty

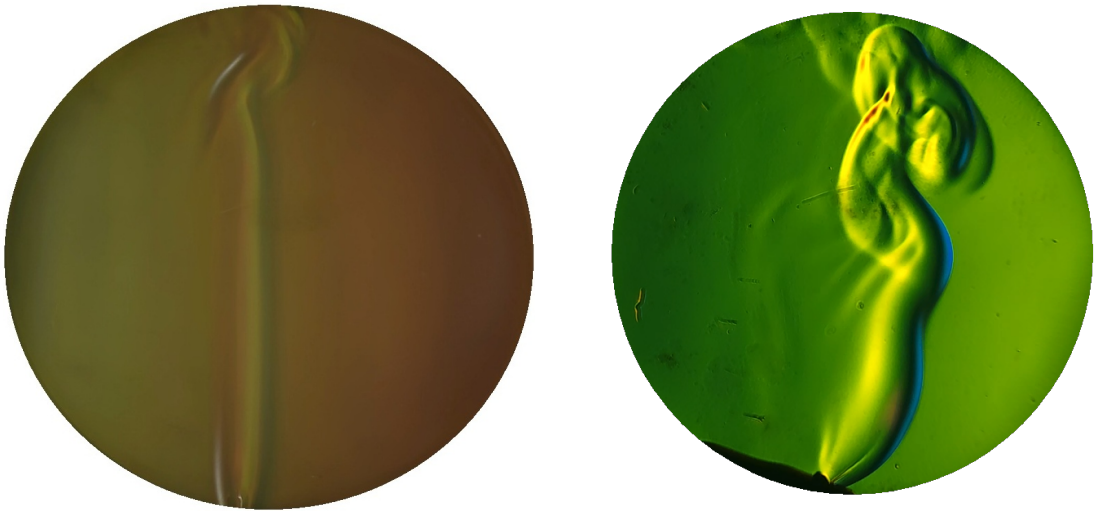
Table 3.8: Measurement uncertainty sources.

Source	Description	Estimate
Regulator gauge pressure	Bourdon gauge accuracy	±0.2 bar
Ambient pressure	Barometric reading, Kathmandu	±0.5 hPa
Thermistor reading	Calibration uncertainty	±1-2°C
Gladstone-Dale constant	Literature value for air	±0.5%
Pixel size	$\Delta x = 0.278$ mm/px	±2%

Chapter 4: RESULTS AND DISCUSSION

4.1 Schlieren System Validation and Filter Characterisation

The Z-type double-mirror schlieren system was validated progressively first on a candle plume to confirm basic optical alignment and sensitivity, then on natural convection from the heated vertical plate, and finally on the wall-jet cases. The single-mirror arrangement was discarded because it produced double image artefacts due to off axis geometry. Double mirror Z schlieren system, which produces significantly better results, resolves this by ensuring all the components operate on same optical axis. Digitally printed continuous-gradient filters on transparent sheets proved to be optically inadequate. The dot structure of the color laser printer was visible at the scale of focal spot size. It also severely degraded transmissivity consistent with the findings of [5]. Filters subsequently produced on E6 slide film at a professional photo lab produced smooth, monotonic calibration curve and no visual banding in the obtained images. This improved results is due to much finer grain size of the film and dye based process involved. Figure 4.1 illustrates the qualitative improvement.



(a) image from digitally printed filter (b) image from E6 slide-film filter

Figure 4.1: Schlieren images illustrating the effect of filter fabrication medium.

A systematic comparison across filter widths (1.5 mm to 5 mm in 0.5 mm steps) confirmed expected sensitivity vs dynamic range trade-off. Filters with widths of 1.5 mm

and 2 mm provide the highest hue sensitivity (largest hue change per unit ray deflection) but saturate readily in the very high gradient near wall region, limiting their utility for quantitative extraction. Filters of 2.5 mm and 3 mm width offer a better balance between sensitivity and dynamic range for the present wall jet conditions and were used for all wall jet quantitative measurements in this study. The rectangular slit source aperture provided noticeably stronger schlieren contrast than a circular pinhole of comparable area, consistent with the directional sensitivity advantage of slit geometry reported by [5]. The slit illuminates a larger portion of the parabolic mirror in the direction sensitive to the density gradient, increasing the deflection signal on the filter.

4.2 Nozzle Jet Characterization

Schlieren images of the free jet at 1.0 bar gauge pressure and 1.4 bar gauge pressure captured weak signatures of density gradient with no visible shock structures. This is consistent with the subsonic exit conditions estimated from the CFD plumbing simulation which estimated $M_{\text{exit}} \approx 0.75$ at 1.0 bar and $M_{\text{exit}} \approx 0.9$ at 1.4 bar after accounting for the approximately 30% total-pressure loss along the supply path (Section 3.1.1). The faintness of the density gradient signatures at these conditions also suggests the moderate density contrast available at high subsonic Mach numbers without wall heating. Quantitative RSD measurements were focused on 1.0 bar and 1.4 bar gauge to study subsonic compressible wall jets.

Direct velocity measurement was not possible with the available instruments. The differential Pitot probe is rated to approximately 100 m/s and the hot-wire anemometer to 35 m/s, both well below the CFD estimated exit velocities of 260 m/s to 305 m/s. Temperature measurement at the exit was done with the help of a temperature sensor present in a hot wire anemometer, which showed the exit temperature around 2 degrees Celsius less than the ambient temperature.

4.2.1 Ensemble Convergence

Figure 4.2 shows the RCoV as a function of frame number for all four wall-jet conditions. The cases exhibit a sharp transient spike in the first 50 to 100 frames reaching RCoV values of approximately 0.17 to 0.18, corresponding to the jet establishment period after the valve is opened. The RCoV drops rapidly below the 5% threshold once the flow reaches a stationary state and remains there for the remainder of the acquisition.

In all four cases, the RCoV stabilizes well below the threshold for the majority of the acquisition, confirming that the mean displacement fields are statistically converged. Frames preceding the first sustained drop below 5% stationary threshold are excluded from the mean.

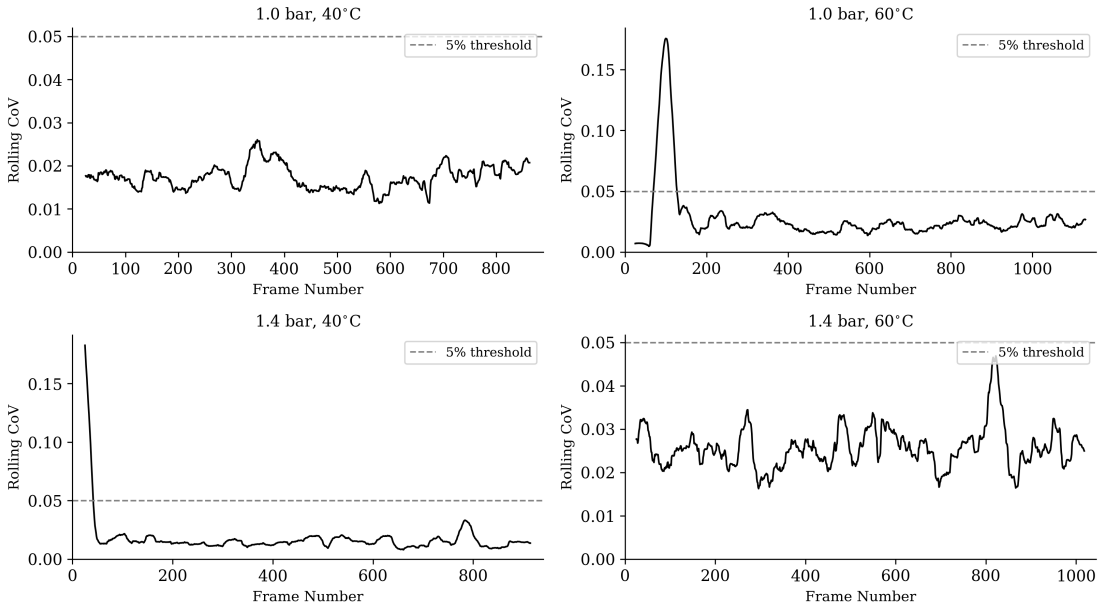
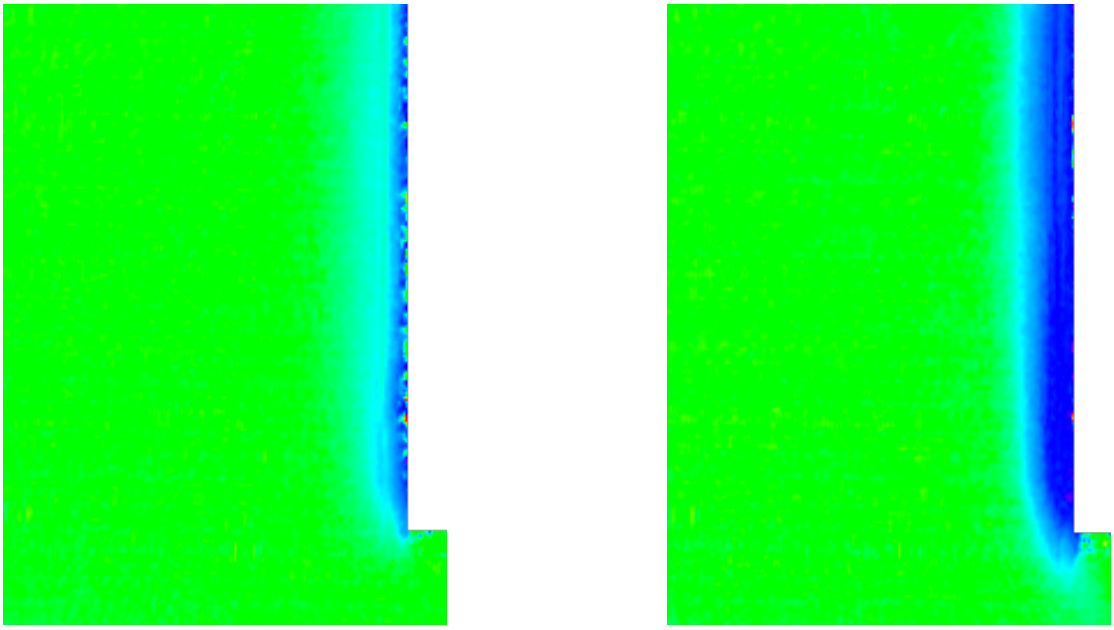


Figure 4.2: Rolling coefficient of variation (RCoV) of the mean absolute displacement.

4.3 Natural Convection Validation

4.3.1 Raw Hue Field and Mean Density Gradient

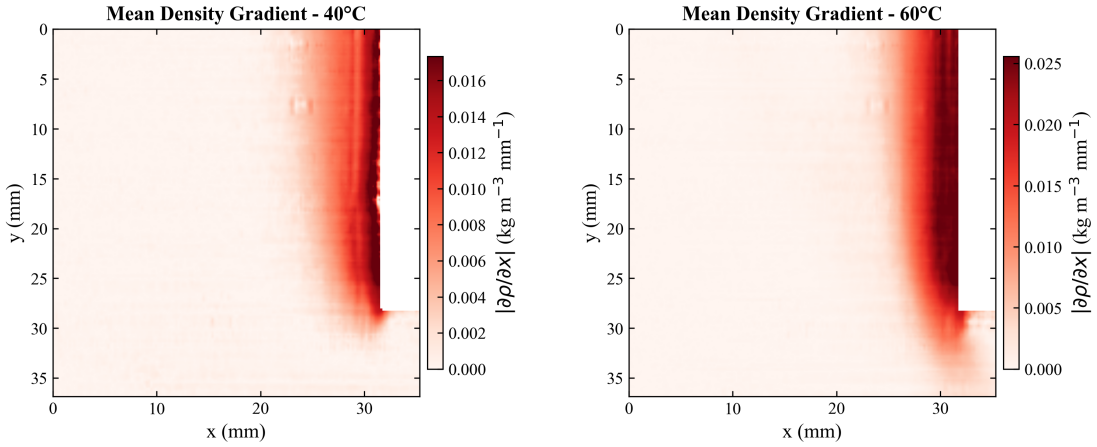
Representative hue-extracted snapshots of the natural convection boundary layer at $T_w = 40°\text{C}$ and $T_w = 60°\text{C}$ are shown in Fig. 4.3, alongside the corresponding mean density gradient fields in Fig. 4.4. In both snapshots, the boundary layer is visible as a narrow cyan to blue hue band adjacent to the right side plate surface, standing out sharply against the ambient green hue. At the higher wall temperature, the blue region is visibly broader, accordant with a thicker thermal boundary layer.



(a) $T_w = 40°\mathrm{C}$ (b) $T_w = 60°\mathrm{C}$

Figure 4.3: Hue extracted RSD snapshots of the natural convection boundary layer.

The mean density gradient magnitude $|\partial\rho/\partial x|$, obtained by averaging approximately 500 frames using 3.5 mm color filter, is shown for both conditions in Fig. 4.4. At $T_w = 40°\mathrm{C}$ the peak gradient reaches approximately 0.016 kg m^{-3} mm^{-1}, while at $T_w = 60°\mathrm{C}$ it rises to roughly 0.026 kg m^{-3} mm^{-1}, scaling in accordance to the larger density difference imposed by the higher wall temperature. In both fields the gradient magnitude falls off monotonically with increasing wall normal distance. The ambient region far from the plate has a near zero gradient in both cases.



(a) $T_w = 40°\mathrm{C}$ (b) $T_w = 60°\mathrm{C}$

Figure 4.4: Mean density gradient magnitude for natural convection from the heated vertical plate.

4.3.2 Reconstructed Density and Temperature Fields

The mean density and temperature fields reconstructed by one dimensional integration of $\partial\rho/\partial x$ under the isobaric assumption are shown in Figs. 4.5 and 4.6 for $T_w = 40°\text{C}$ and $T_w = 60°\text{C}$, respectively. In the density field, the near wall fluid is depleted to roughly 0.99 $\text{kg}\,\text{m}^{-3}$ at 40°C and to approximately 0.94 $\text{kg}\,\text{m}^{-3}$ at 60°C. The corresponding temperature fields reveal a narrow elevated temperature layer near the plate surface, with peak mean temperatures reaching approximately 312 K at 40°C (313 K) and 330 K at 60°C (333 K), both in agreement with the set wall conditions.

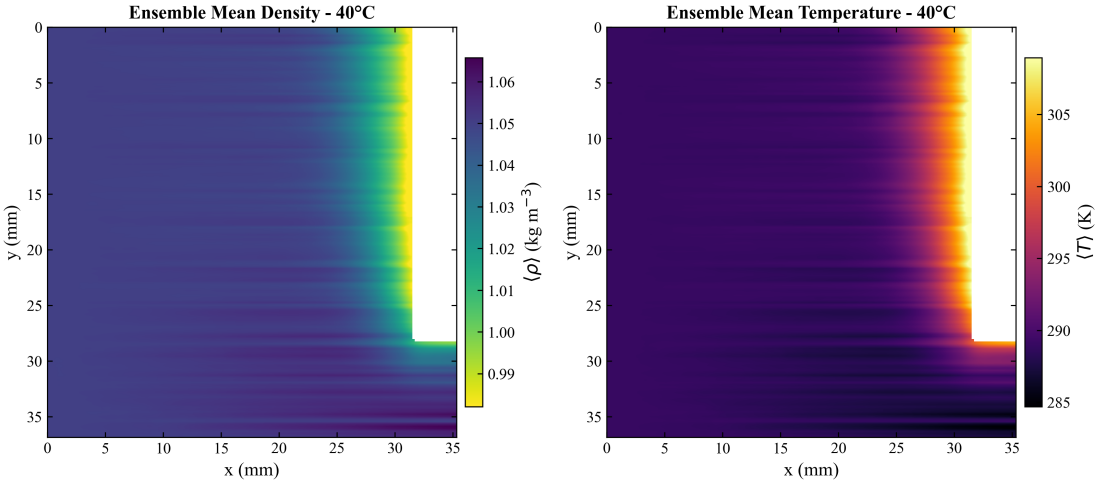


Figure 4.5: mean density and temperature fields reconstructed from the RSD measurement at $T_w = 40°\text{C}$.

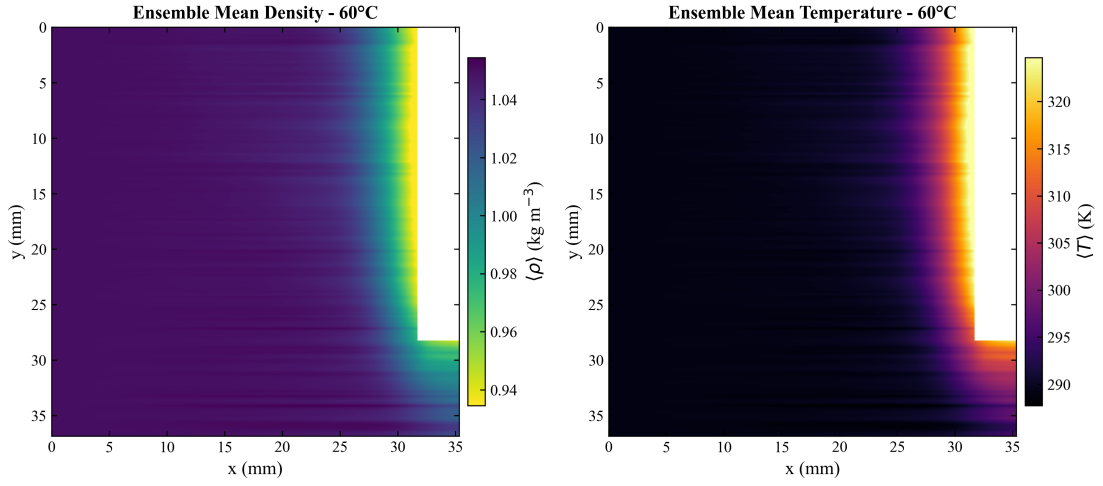


Figure 4.6: mean density and temperature fields reconstructed from the RSD measurement at $T_w = 60°\text{C}$.

4.3.3 Wall Normal Profiles and Wall Temperature Validation

Wall normal density and temperature profiles extracted at three representative heights along the plate ($y = 3$, 11, and 18 mm from top) are presented in Fig. 4.7 for $T_w = 40°C$ and in Fig. 4.8 for $T_w = 60°C$. At all heights, the density profiles decay smoothly from the ambient reference ρ_0 toward a minimum at the plate surface, while the temperature profiles rise correspondingly from T_0 to a near wall maximum. The three height levels collapse almost onto a single curve in the undisturbed ambient region, differing only within the thermal boundary layer close to the wall. The profiles at higher y (closer to the leading edge of the plate) show marginally sharper gradients. This explains the expected thinning of the laminar boundary layer toward the leading edge.

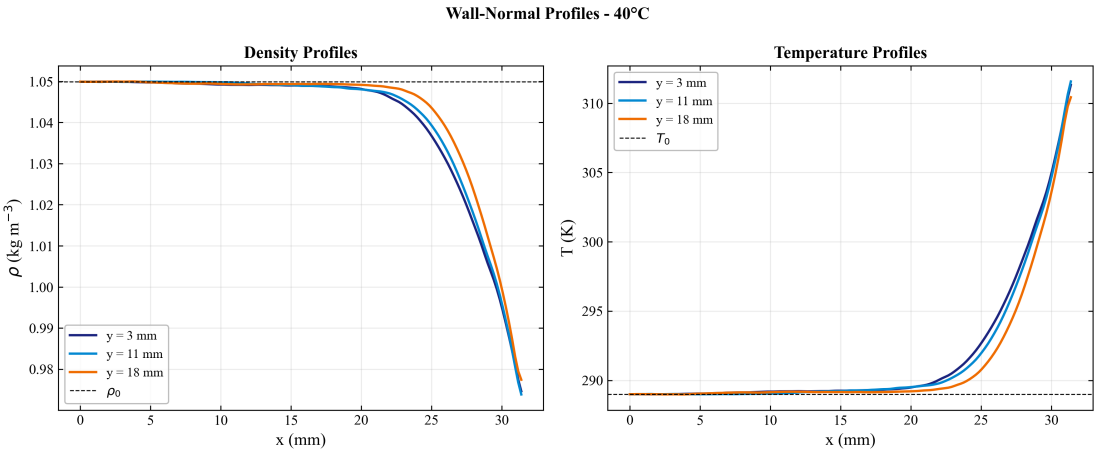


Figure 4.7: Wall normal density and temperature profiles at $y = 3$, 11, and 18 mm for $T_w = 40°C$.

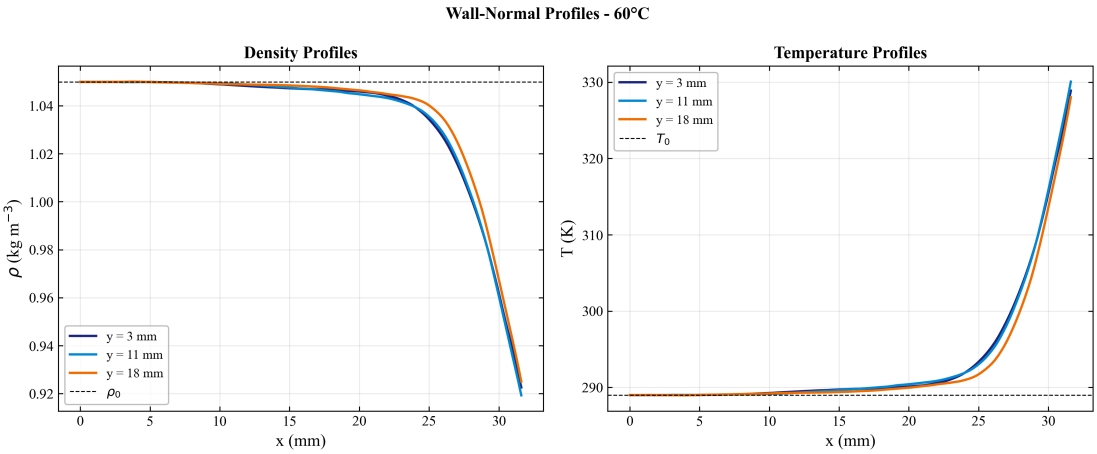


Figure 4.8: Wall normal density and temperature profiles at $y = 3$, 11, and 18 mm for $T_w = 60°C$.

The primary validation metric is the wall temperature recovered by extrapolating the reconstructed temperature profile to $x = x_{\text{wall}}$. At $T_w = 40°\text{C}$ the profiles reach approximately 312 K at the plate surface, compared with the thermocouple set-point of 313 K, an agreement within $\sim 0.3\%$. At $T_w = 60°\text{C}$, the extrapolated wall temperature reaches approximately 330 K against the set point of 333 K, a discrepancy of roughly 1%. Across both conditions the RSD experiments recover the imposed wall temperature to within experimental uncertainty, validating the end to end measurement and reconstruction procedure.

4.4 Wall Jet Results

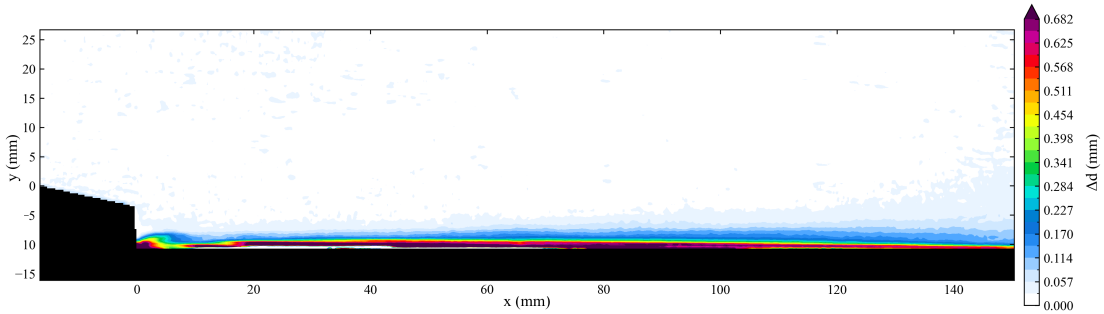
The two operating conditions correspond to nozzle exit velocities of approximately 260 m/s ($M \approx 0.75$) at 1.0 bar gauge regulator pressure and 305 m/s ($M \approx 0.9$) at 1.4 bar gauge regulator pressure, as estimated from CFD simulation of the full supply plumbing. Both conditions are high-subsonic compressible flows. Neither exhibits shock structures in the schlieren imagery, consistent with subsonic exit. The difference in exit Mach number between the two cases ($\Delta M \approx 0.15$) means that compressibility effects contribute to density-field differences between pressure conditions in addition to thermal boundary layer effects.

4.4.1 Mean Displacement Field

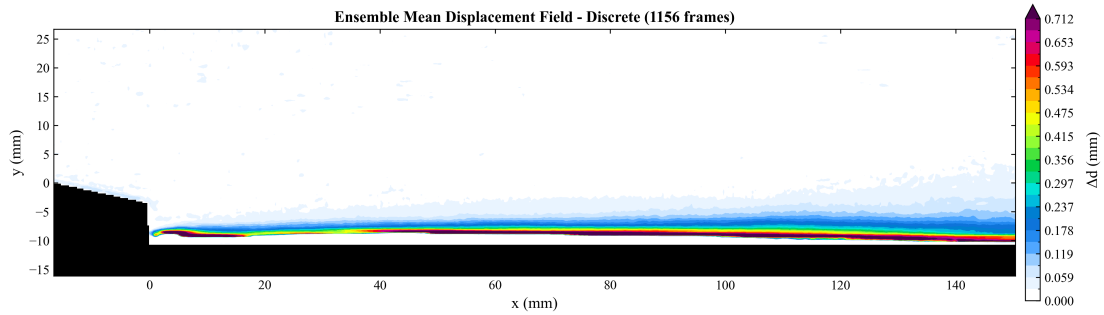
The mean displacement fields for all four wall jet conditions are shown in Figs. 4.9 and 4.10. Several features distinguish these fields from the natural convection case. First, the high displacement band is substantially thinner and more intense near the nozzle exit, reflecting the dominance of forced convection in compressing the thermal layer into a narrower wall normal extent. Second, the band extends further in the streamwise direction before decaying to the ambient level, consistent with the higher convective velocity sustaining the thermal gradient over a longer development length. Third, the displacement magnitude is larger at the higher wall temperature for both pressure conditions, as expected from the stronger wall to ambient temperature difference driving the density gradient.

At the higher pressure (1.4 bar, Fig. 4.10), the near-wall displacement band is visibly more compact in the wall normal direction compared with the 1.0 bar case at the same wall temperature. This compression of the displacement layer is consistent with the

thinning of the thermal boundary layer under stronger jet momentum, and is observed at both wall temperature settings.

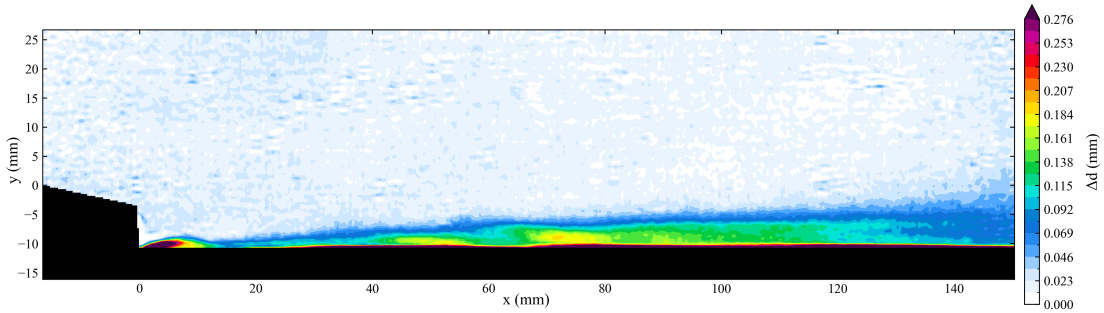


(a) $p_{\text{gauge}} = 1.0$ bar, $T_w = 40°\text{C}$

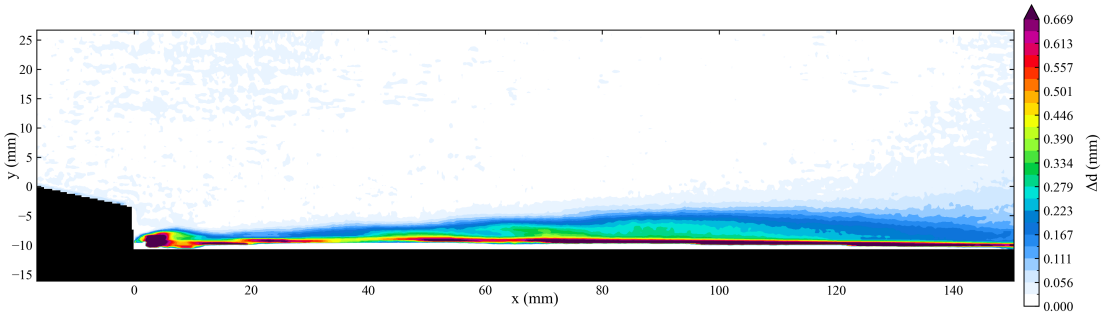


(b) $p_{\text{gauge}} = 1.0$ bar, $T_w = 60°\text{C}$

Figure 4.9: mean displacement fields for the wall jet at 1.0 bar gauge.



(a) $p_{\text{gauge}} = 1.4$ bar, $T_w = 40°\text{C}$

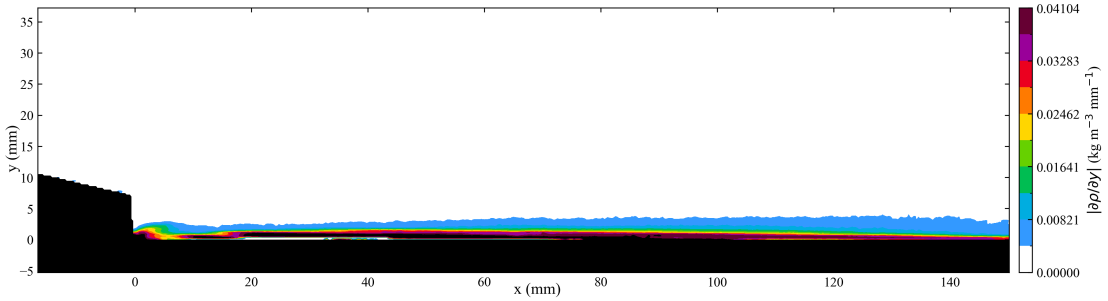


(b) $p_{\text{gauge}} = 1.4$ bar, $T_w = 60°\text{C}$

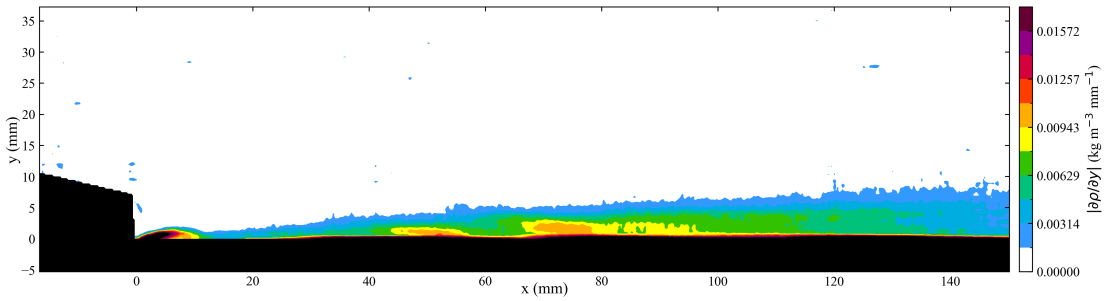
Figure 4.10: mean displacement fields for the wall jet at 1.4 bar gauge.

4.4.2 Mean Density Gradient Field

The mean density gradient fields $|\partial\rho/\partial y|$ for all four conditions are presented in Figs. 4.11 and 4.12. In every case, a well defined high gradient band is present immediately adjacent to the plate surface, corresponding to the coupled thermal and shear layer of the wall jet. This band is substantially thinner and more intense than the one observed in the natural convection case at the same wall temperature, which had a peak value of $\approx 0.028\ \mathrm{kg\,m^{-3}\,mm^{-1}}$ over a width of several millimeters. The dominant role of forced convection in compressing the thermal layer is evident here. At $T_w = 40°\mathrm{C}$ (Fig. 4.11), the near wall gradient band is measurable but relatively thin and low in peak magnitude. The gradient decays in the streamwise direction as the jet decelerates and the shear layer grows. The 1.4 bar case produces a noticeably more intense and narrower band than the 1.0 bar case at the same wall temperature, consistent with the stronger forced convection suppressing the thermal layer thickness. At $T_w = 60°\mathrm{C}$ (Fig. 4.12), the contrast between pressure conditions is stronger. The higher wall temperature drives a stronger density difference, and the interaction with jet momentum produces a more pronounced variation in near wall gradient structure between the two pressure cases.

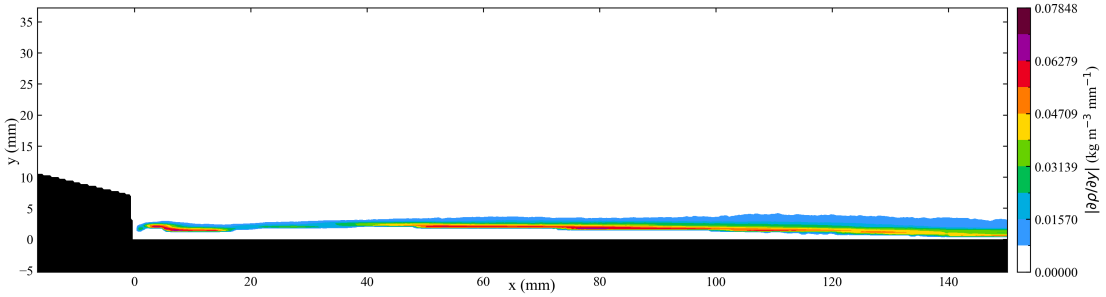


(a) $p_{\mathrm{gauge}} = 1.0$ bar, $T_w = 40°\mathrm{C}$

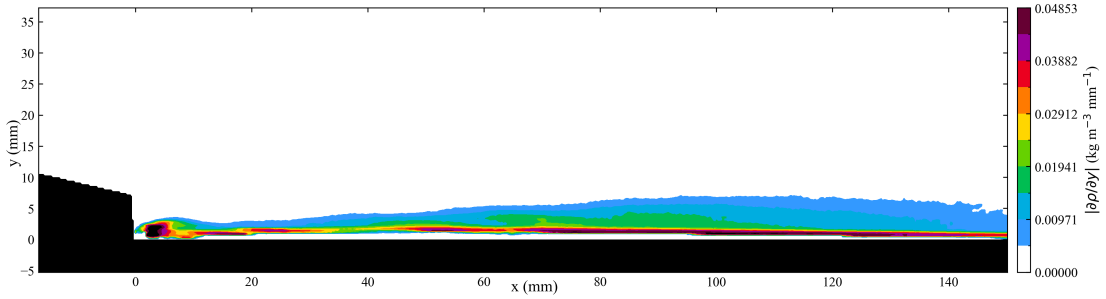


(b) $p_{\mathrm{gauge}} = 1.4$ bar, $T_w = 40°\mathrm{C}$

Figure 4.11: mean density gradient fields for the wall jet at $T_w = 40°\mathrm{C}$.



(a) $p_{\text{gauge}} = 1.0$ bar, $T_w = 60°\text{C}$



(b) $p_{\text{gauge}} = 1.4$ bar, $T_w = 60°\text{C}$

Figure 4.12: mean density gradient fields for the wall jet at $T_w = 60°\text{C}$.

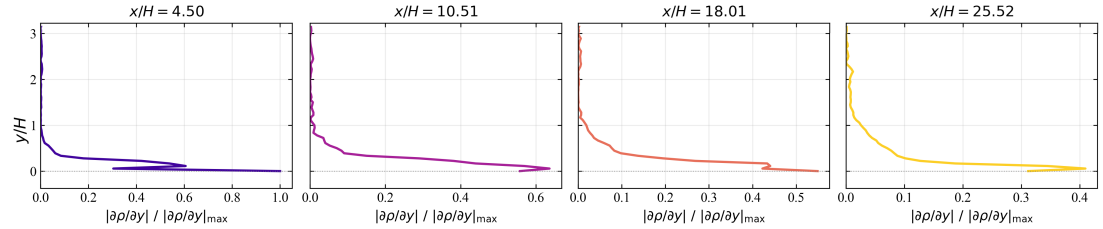
4.4.3 Wall Normal Gradient Profiles

Wall normal profiles of the normalized density gradient $|\partial\rho/\partial y| \,/\, |\partial\rho/\partial y|_{\max}$ extracted at four streamwise positions ($x/H \approx 4.5$, 10.5, 18.0, and 25.5, where $H = 5$ mm is the nozzle exit height) are shown in Fig. 4.13 for all four conditions. Each profile collapses the wall-normal coordinate as y/H and is normalized by the local peak gradient, enabling direct comparison of the boundary-layer shape across conditions while the absolute magnitudes are compared in Fig. 4.14.

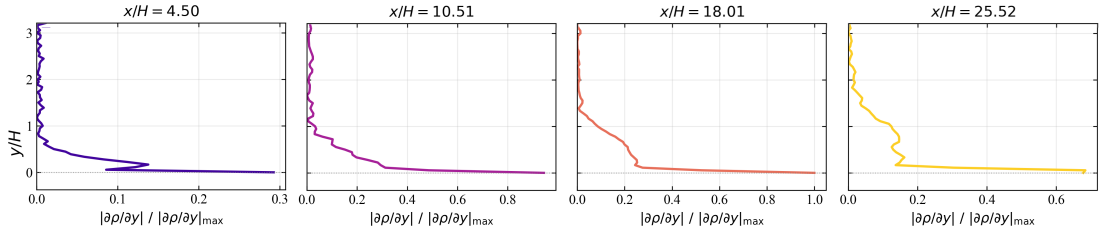
Two features are common to all conditions. First, the peak gradient migrates away from the wall and the profile broadens with increasing x, consistent with the streamwise growth of the thermal boundary layer in a developing wall jet. Secondly, a two region structure is visible in most profiles. A near wall region of elevated gradient associated with the thermal boundary layer, and a relatively flat, low gradient region extending into the outer flow. The transition between these regions sharpens at higher jet momentum (1.4 bar cases), where the thermal layer is more tightly confined.

At 1.4 bar (Fig. 4.13b,d), the near wall gradient peak is sharper and more concentrated at small y/H than the corresponding 1.0 bar profiles. This compression of the gradient layer under stronger forced convection is consistent with a thinner momentum boundary

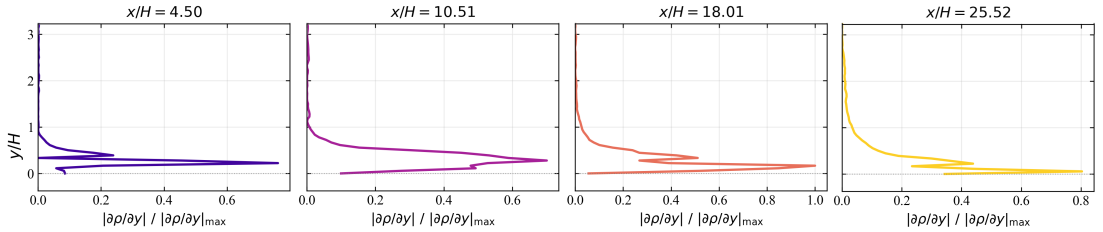
layer suppressing thermal diffusion toward the wall. At 1.0 bar, the profiles are broader and the peak is less pronounced, indicating a thicker and more gradual thermal boundary layer.



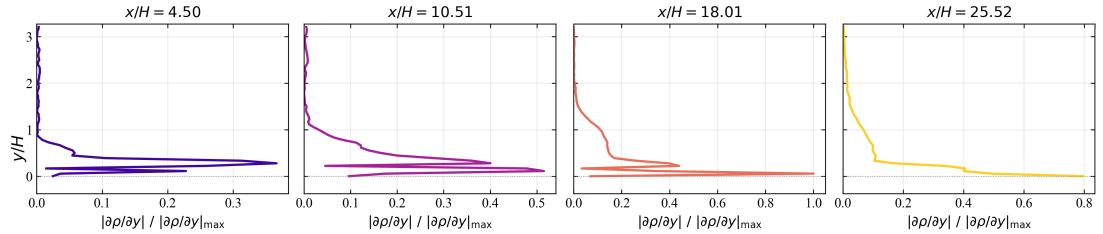
(a) $p_{\text{gauge}} = 1.0$ bar, $T_w = 40°\text{C}$



(b) $p_{\text{gauge}} = 1.4$ bar, $T_w = 40°\text{C}$



(c) $p_{\text{gauge}} = 1.0$ bar, $T_w = 60°\text{C}$



(d) $p_{\text{gauge}} = 1.4$ bar, $T_w = 60°\text{C}$

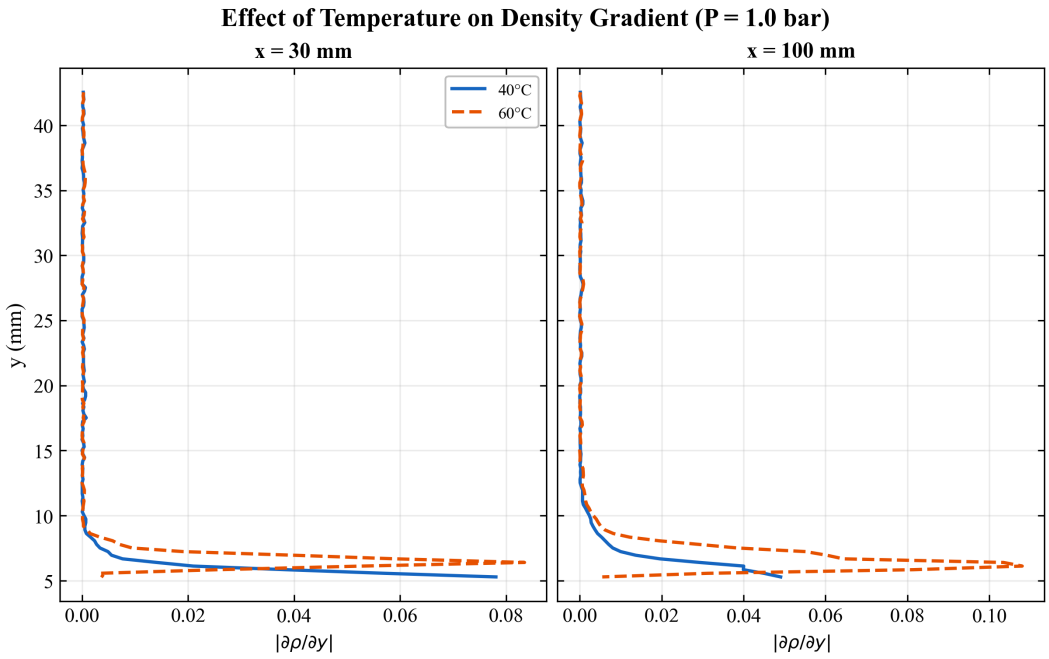
Figure 4.13: Wall normal normalized density-gradient profiles at four streamwise stations.

4.4.4 Effect of Wall Temperature

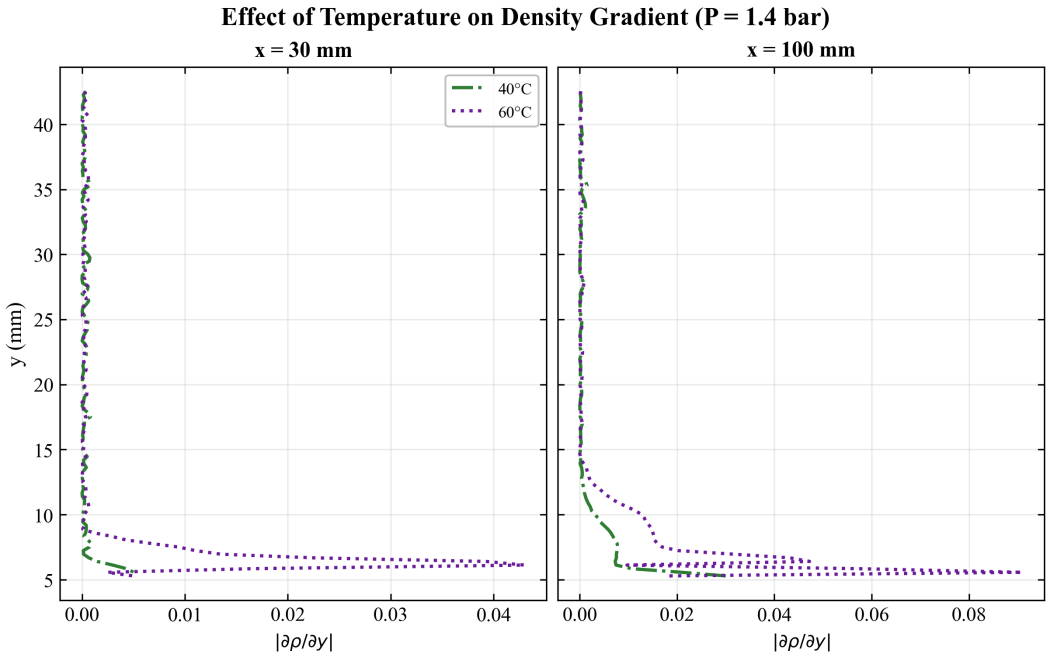
Figure 4.14 compares wall normal density gradient profiles at $x = 30$ mm and $x = 100$ mm for wall temperatures of 40°C and 60°C at each operating pressure. At $p_{\text{gauge}} = 1.0$ bar, a clear separation exists between the two temperature cases at both measurement stations. The 60°C case produces notably larger near wall gradient magnitudes, and the separation

intensifies with streamwise distance, reflecting progressive thermal boundary layer development along the heated plate. The higher wall temperature also produces a broader near wall gradient profile, which agrees with the steeper transverse density variation due to higher wall to fluid temperature difference. At $p_{\text{gauge}} = 1.4$ bar, the temperature effect is comparatively subdued. At $x = 30$ mm the two temperature cases are nearly indistinguishable, indicating that the increased convective transport at higher jet momentum partially offsets the influence of elevated wall temperature on the near wall density gradient structure. At $x = 100$ mm, a separation between the profiles becomes apparent, though less striking than in the 1.0 bar case. This attenuation is physically accordant with increased jet momentum enhancing convective heat removal from the near wall region. Although the wall temperature boundary condition is identical, the stronger convective flux reduces the near-wall temperature gradient and therefore the density gradient.

The wall temperature is thus found to scale the near wall gradient magnitude without fundamentally altering the jet development length scales. That is, the streamwise decay rate of the peak gradient is similar at both temperatures within each pressure case.



(a) $p_{\text{gauge}} = 1.0$ bar. A clear distinction exists between $T_w = 40°\text{C}$ and $60°\text{C}$ at both stations and grows with x.



(b) $p_{\text{gauge}} = 1.4$ bar. The temperature effect is suppressed in the near field ($x = 30$ mm) by stronger jet momentum and becomes apparent only further downstream ($x = 100$ mm).

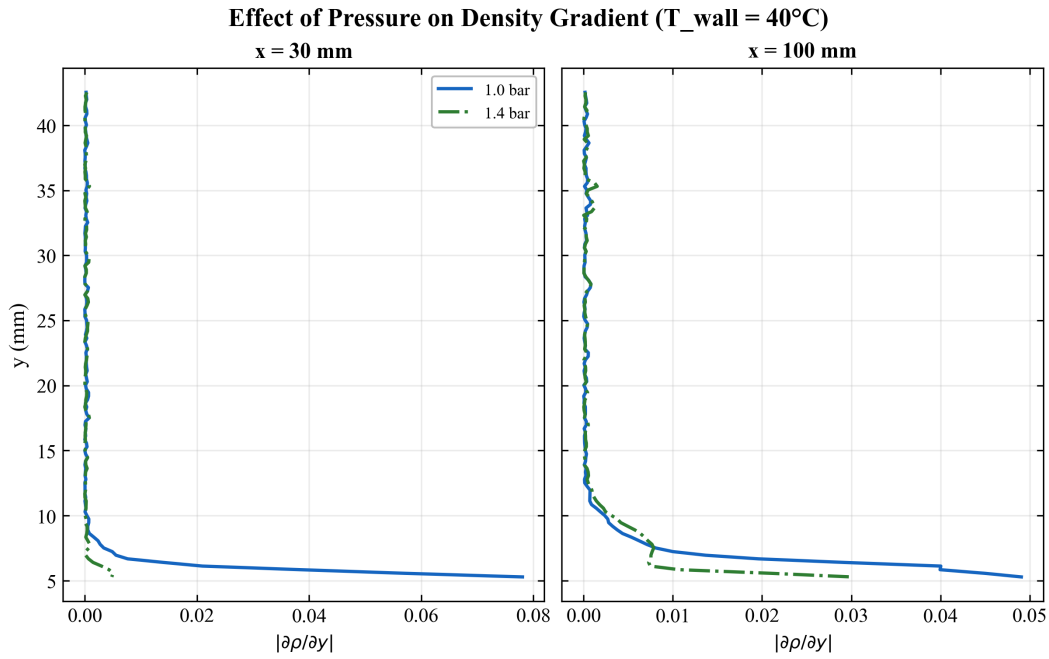
Figure 4.14: Effect of wall temperature on the wall normal density gradient profiles at two streamwise stations and two operating pressures. Each plot shows $|\partial\rho/\partial y|$ versus wall normal distance y.

4.4.5 Effect of Reservoir Pressure

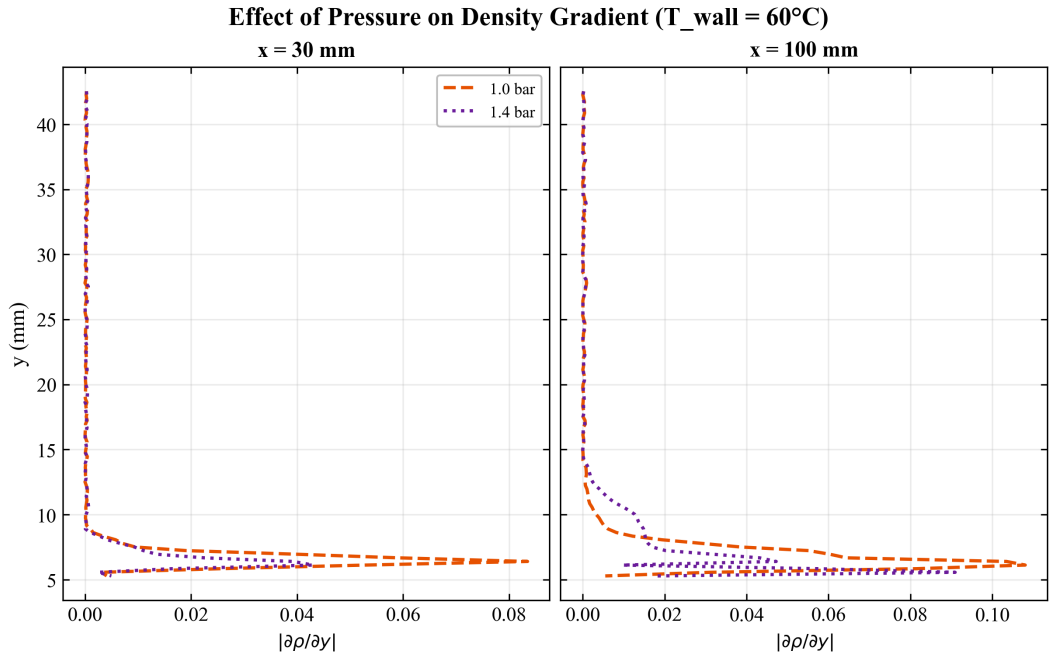
Figure 4.15 compares wall-normal density-gradient profiles at the same two streamwise stations for 1.0 bar and 1.4 bar gauge at each wall temperature. At $T_w = 40°\text{C}$, the two pressure cases produce nearly identical profiles at $x = 30$ mm, with only a modest divergence appearing further downstream at $x = 100$ mm. This suggests that at a moderate wall-to-ambient temperature difference, the thermal boundary layer development is relatively insensitive to the change in jet momentum flux in the near-field. At $T_w = 60°\text{C}$, the effect of pressure becomes considerably more pronounced, particularly at $x = 100$ mm, where the 1.0 bar case exhibits significantly larger near wall density gradient magnitudes than the 1.4 bar case.

This behavior can be attributed to the coupling between thermal effects and jet momentum. At lower jet momentum (1.0 bar), wall heating sustains a thicker and more intense thermal boundary layer. At higher jet momentum (1.4 bar), the increased convective transport suppresses the near wall density gradient despite the same wall temperature boundary condition. The momentum effect is thus strongly coupled to the wall temperature, becoming physically significant only when the wall to ambient temperature difference is sufficiently large to produce thermal effects that are competitive with the jet momentum.

The contribution of compressibility should also be noted. The difference in exit Mach number ($\Delta M \approx 0.15$) between the two pressure conditions means that the density field is intrinsically different between cases even in the absence of wall heating. The 1.4 bar case carries a higher than ambient density at the jet exit, which adds to the thermal density gradient in the near wall region.



(a) $T_w = 40°C$, momentum effects are less striking at this wall temperature, with the two cases converging in the near field.



(b) $T_w = 60°C$, momentum effects are more striking, at $x = 100$ mm the 1.0 bar case produces larger near wall gradients, indicating stronger buoyancy momentum coupling.

Figure 4.15: Effect of reservoir pressure on the wall normal density gradient profiles at two streamwise stations and two wall temperatures.

4.4.6 Streamwise Decay of Peak Density Gradient

The streamwise evolution of the peak near wall density gradient for all four conditions are summarized in Figs. 4.16 and 4.17. Across all cases the peak gradient decays monotonically in the streamwise direction following a rapid initial adjustment in the first 10 mm to 20 mm downstream of the nozzle exit. Both increasing wall temperature and increasing reservoir pressure elevate the peak gradient level, while the shape of the decay curve remains broadly similar. It suggests that the underlying jet development is not fundamentally altered by either parameter within the tested range. At $T_w = 40°C$ (Fig. 4.16), the 1.0 bar case shows consistently higher peak gradients than the 1.4 bar case. It is attributed to the different compressibility contributions at the two Mach numbers. At 1.4 bar, the stronger forced convection removes heat more efficiently reducing the near wall temperature gradient faster than the increased Mach number can compensate. At $T_w = 60°C$ (Fig. 4.17), the difference is less prominent. This can be attributed to temperature of the plate dominating flow more significantly near walls than at $T_w = 40°C$, making the two reservoir gauge conditions indifferent in terms of density gradient decay.

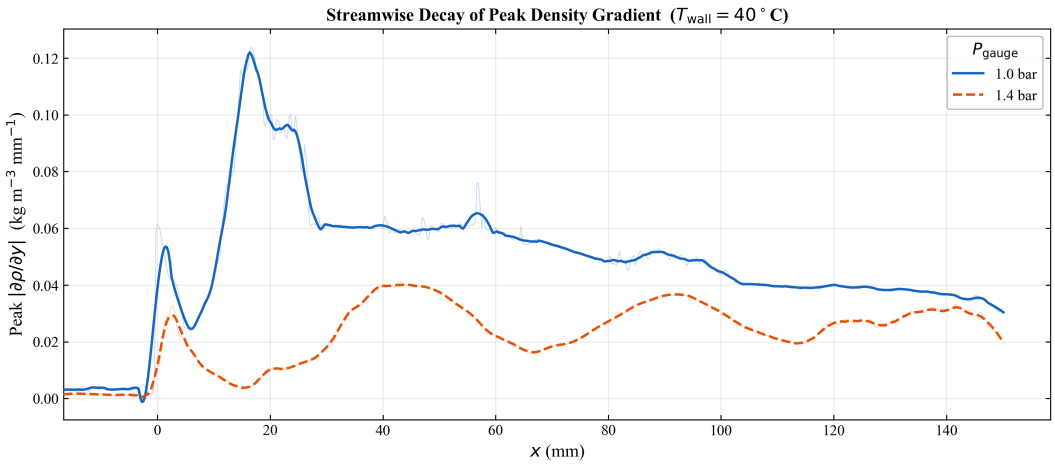


Figure 4.16: Streamwise decay of the peak mean density gradient at $T_w = 40°C$ for both operating pressures.

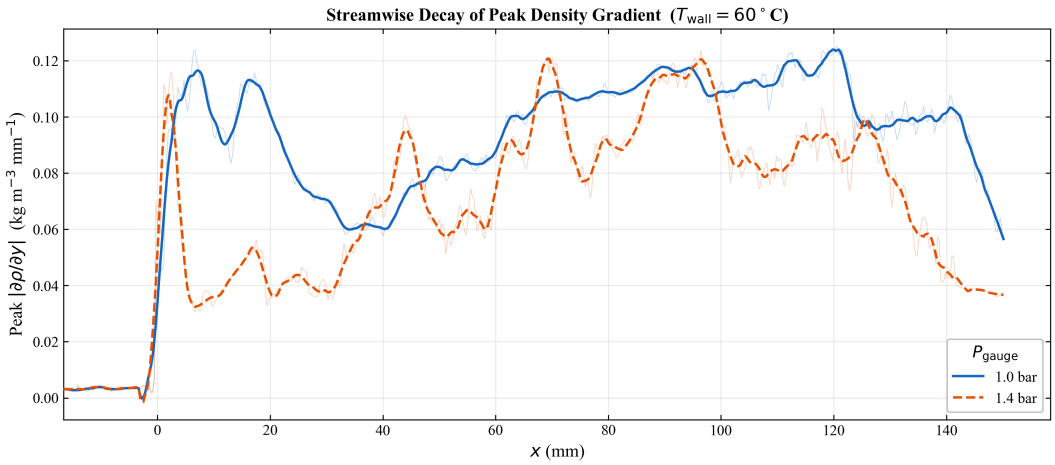


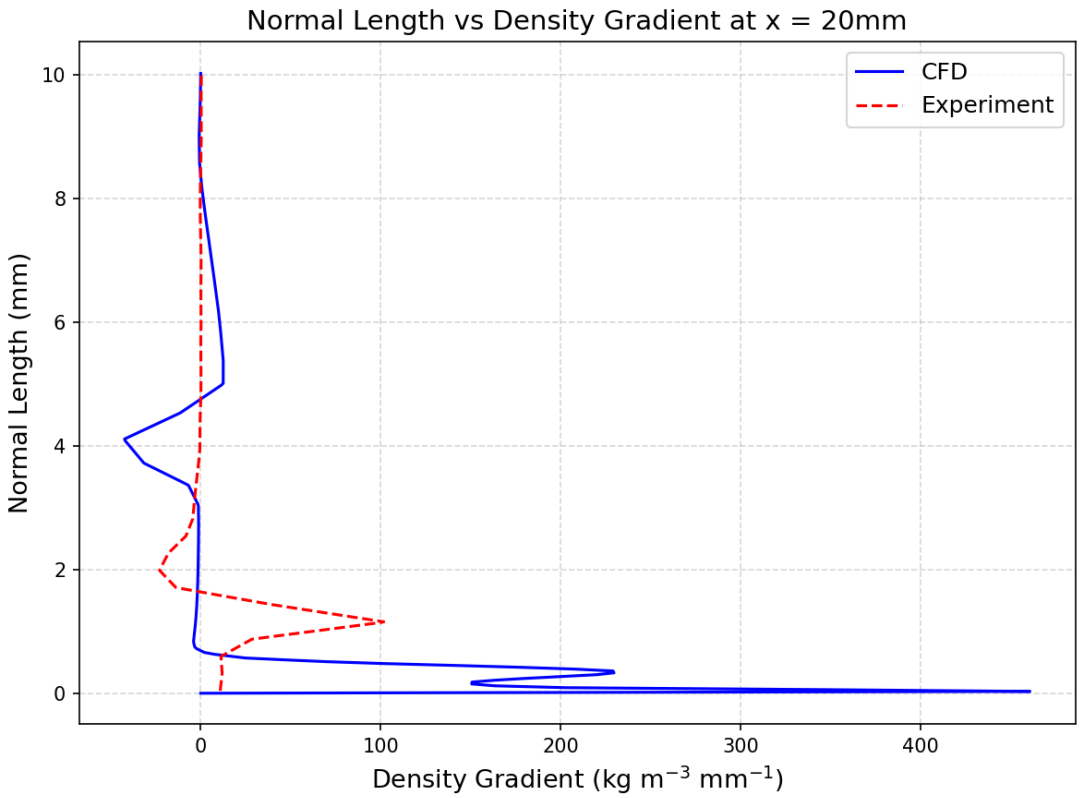
Figure 4.17: Streamwise decay of the peak mean density gradient at $T_w = 60°$C for both operating pressures. The elevated wall temperature increases peak gradient magnitudes at both pressures.

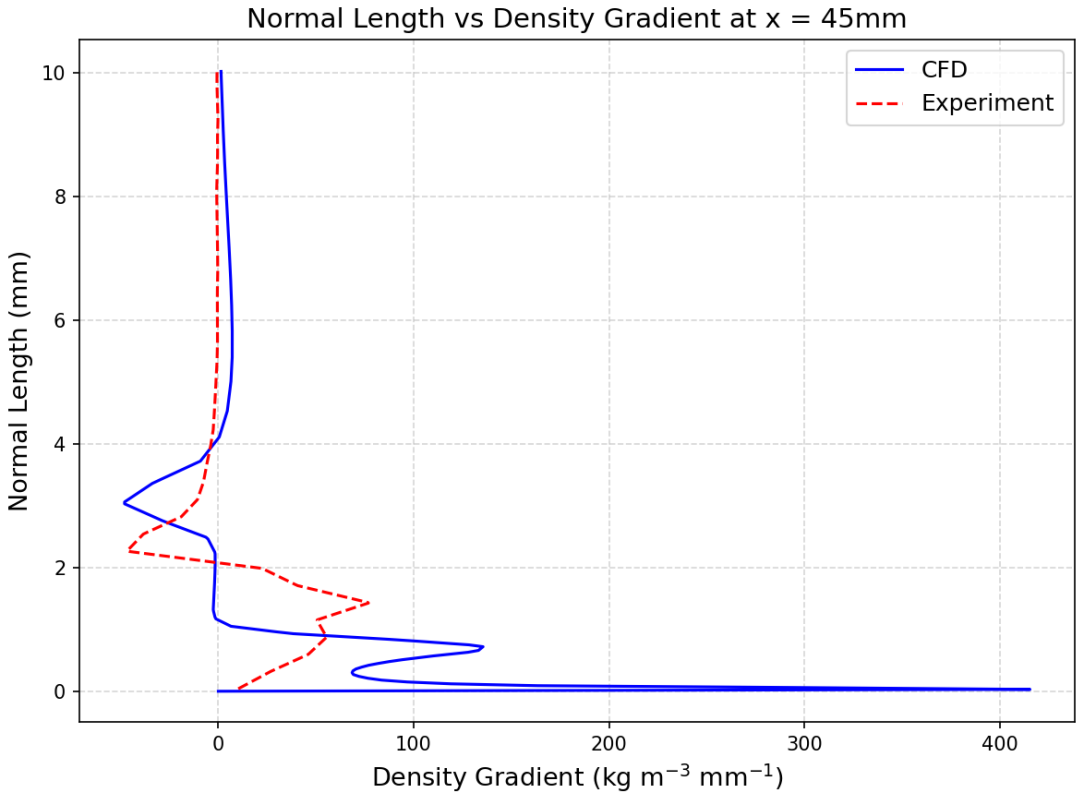
4.5 CFD Results and Comparison with Experiment

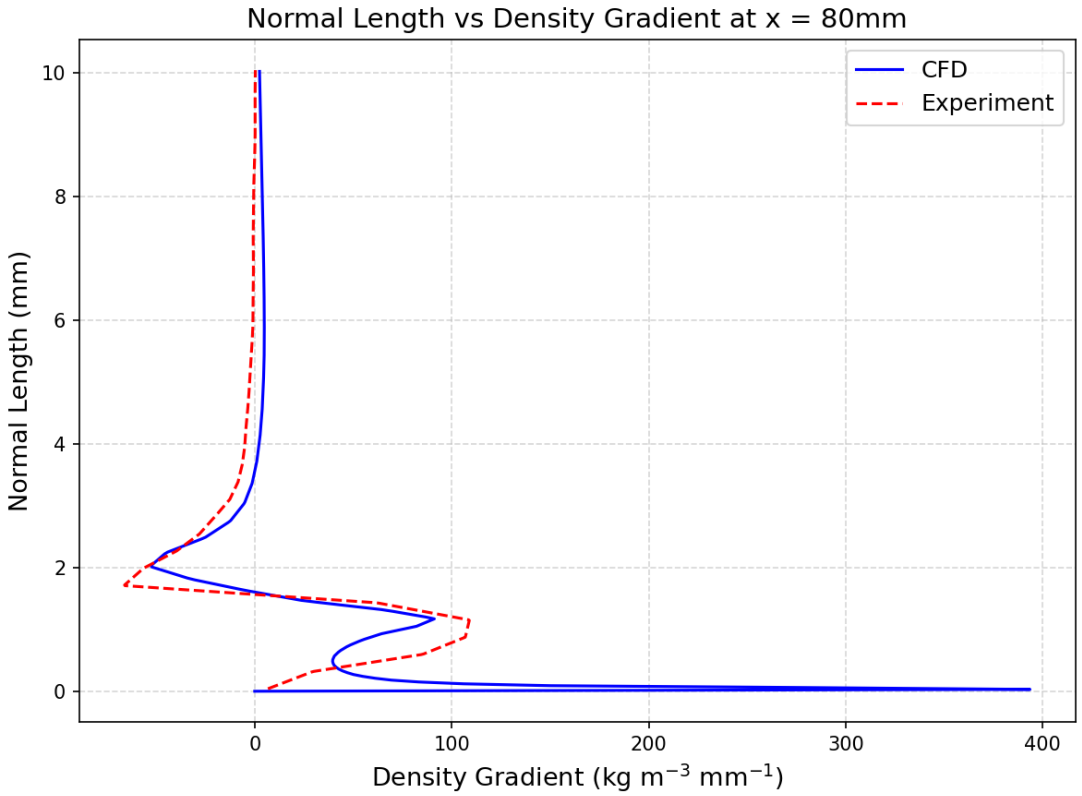
Wall-normal profiles of the density gradient magnitude from the 2D RANS simulation are compared with the RSD derived experimental profiles at four streamwise stations ($x = 20$, 45, 80, and 120 mm) for the condition $p_{\text{gauge}} = 1.0$ bar, $T_w = 60°$C in Fig. 4.18. Both the CFD and experimental profiles show a near wall gradient peak that decays in magnitude and shifts away from the wall with increasing x. However, quantitative differences are evident at all stations. The CFD peak gradient magnitude exceeds the experimental value at every station. At $x = 20$ mm the CFD peak is approximately an order of magnitude larger than the RSD measurement. This discrepancy decreases with downstream distance but is present even at $x = 120$ mm. Conversely, the experimental profiles show a more diffuse near wall gradient region that extends further in the wall normal direction than the CFD prediction. Several factors contribute to these differences. The most immediate is the turbulence modeling approach. The k-ω SST model tends to over predict near wall velocity and temperature gradients in heated wall jet interactions. Although k-ω SST is generally superior to standard k-ε in adverse pressure gradient and separating flows owing to its blended near wall treatment, it remains an eddy viscosity closure and therefore cannot represent the anisotropic Reynolds stresses and turbulent heat flux that govern mixing in a heated compressible shear layer. The result is an overly sharp, intense near wall gradient in the simulation relative to the experiment. The 2D assumption compounds this by compressing all flow physics into a single plane, whereas the experimental jet is inherently three-dimensional. Spanwise variations near the nozzle side edges and secondary flows induced by the rectangular exit geometry produce

cross plane mixing that the simulation cannot capture. This three dimensionality likely accounts for some of the broader, more diffuse gradient profiles seen experimentally. On the measurement side, RSD integrates the density gradient along the full spanwise depth of the test section ($L = 150$ mm), yielding a line-of-sight average. This spatial averaging smooths sharp near wall gradients that the CFD resolves at the continuum level. So, part of the apparent discrepancy reflects a genuine difference in what the two methods measure rather than an error in either. The wall temperature boundary condition provides an additional source of discrepancy. A uniform $T_w = 60^{\circ}$C is imposed in the simulation, whereas the experimental plate temperature is unlikely to be perfectly uniform, particularly near the leading edge where thermal entry effects are developing. Even small differences in the thermal boundary condition can have a noticeable impact on the near wall density gradient. Finally, at a jet velocity of approximately 260 m/s ($M \approx 0.75$), the steady RANS solution cannot capture unsteady density fluctuations associated with turbulent mixing at this Mach number, whereas RSD is directly sensitive to time averaged density variations that include contributions from these fluctuations via the mean. The k ω SST model also uses a Boussinesq eddy viscosity assumption for the turbulent heat flux, which is known to be inaccurate in flows where the principal strain and temperature gradient directions are misaligned, a condition that is very likely in the near nozzle region of a heated wall jet. This is most significant near the nozzle exit where turbulent structures are most energetic, in agreement to the largest discrepancy occurring at $x = 20$ mm.

It can be observed that the experimental results from RSD and CFD results agree consistently in terms of density gradient profile. They follow similar sharp increase in density gradient near wall and sharp decrease in density gradient in the outer layer of jet. The sharp increase in density gradient near wall corresponds to decreasing temperature from hot wall to cold jet. The second peak with decreasing density gradient corresponds to outer shear layer region where it warms up due to kinetic energy dissipating in the form of heat.


Normal Length vs Density Gradient at x = 20mm
CFD
Experiment
Normal Length (mm)
Density Gradient (kg m^{-3} mm^{-1})
0
2
4
6
8
10
0
100
200
300
400


Normal Length vs Density Gradient at x = 45mm
CFD
Experiment
Normal Length (mm)
Density Gradient (kg m^{-3} mm^{-1})
0
2
4
6
8
10
0
100
200
300
400



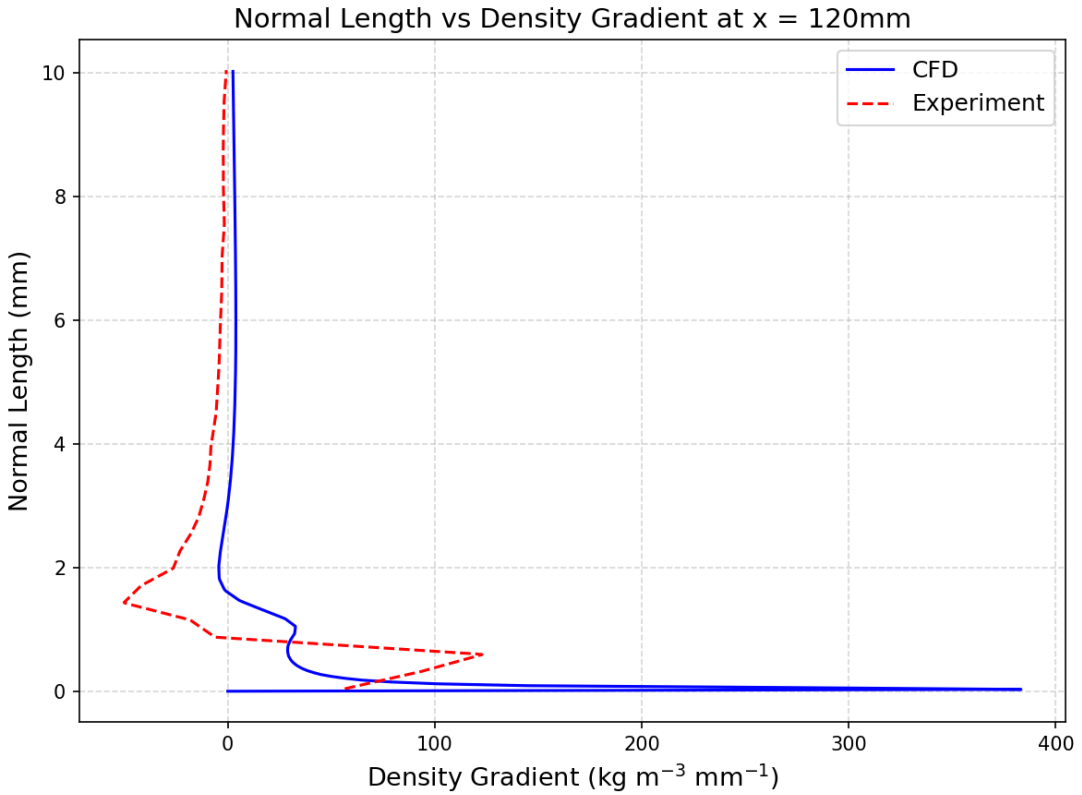


Figure 4.18: Comparison of wall normal density gradient profiles: experimental RSD and 2D RANS CFD at four streamwise stations ($p_{\text{gauge}} = 1.0$ bar, $T_w = 60^{\circ}$C)..

4.6 Qualitative RSD Flow Visualisation

While it has been demonstrated above the quantitative density gradient measurements from RSD, this section demonstrates the qualitative visualization capability of the schlieren system using flows of known character. These images serve two purposes. First, they validate the optical alignment and sensitivity of the Z-type double mirror system. Second, they illustrate the additional information content that color encoding provides relative to conventional knife-edge schlieren, motivating the use of RSD for the qualitative uses as well.

Candle Plume

The candle plume is a classical test case for schlieren systems owing to its strong buoyancy-driven density gradients and experimental accessibility [16]. Figure 4.19 shows a representative RSD image acquired at the present facility.

The flame base appears as a compact red-orange and blue band corresponding to the steepest temperature gradient and largest filter deflection. Moving upward, the rising hot gas column transitions through yellow and green hue as the density gradient weakens. Above the flame tip, the turbulent transition of the plume is visible as an unsteady vortical hue pattern. The green background corresponds to the filter center hue, indicating zero deflection in the undisturbed ambient.

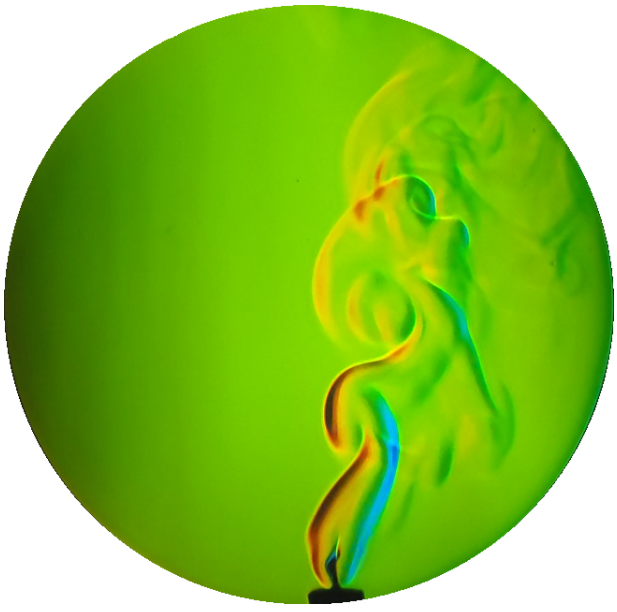
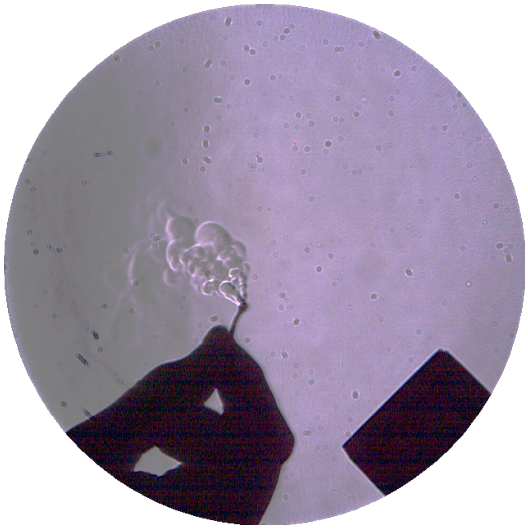


Figure 4.19: RSD image of a candle plume.

Matchstick Ignition: Knife-Edge versus Color Filter

To directly compare conventional knife edge schlieren with RSD, paired image sequences of a matchstick ignition were acquired in separate runs under identical conditions. One run used the rainbow color filter for RSD whereas the other replaced the filter with a knife edge oriented for vertical cutoff. The density gradient is encoded in same one direction in both of these methods. Figure 4.20 shows representative frames from both sequences at a comparable instant during ignition. In the knife-edge image (Fig. 4.20a), the hot gas rising from the match head is visible as a bright and dark structure, and the overall flow pattern is recognizable. However, the image encodes only the sign and relative magnitude of the gradient in one direction. It is quite challenging to infer the extent of change in density gradient from brightness alone. In the RSD image (Fig. 4.20b), the same region is encoded as a continuous hue field. The steep temperature driven density gradients appear as Red-Orange hue farthest from background light blue hue, followed by darker blue and greenish hue. The direction of gradient can also be easily inferred by comparing if the hue is greater or less than background hue. The qualitative aspects of spatial structure of the density gradient field is therefore directly legible from a single RSD frame without requiring additional processing.



(a) Knife-edge schlieren (vertical cutoff). The hot gas plume is visible as a bright and dark structure.

(b) RSD with linear color filter. The flame and plume structures are directly distinguishable by hue in a single frame.

Figure 4.20: Comparison of knife-edge schlieren and RSD for a matchstick ignition, acquired in separate runs under identical conditions.

Flame Propagation in a Butane Plume

A further qualitative demonstration was performed using RSD to visualize a candle flame descending into rising butane gas plumes from a lighter nozzle. Butane has a molecular weight of approximately $58\,\mathrm{g\,mol^{-1}}$, roughly twice that of air ($29\,\mathrm{g\,mol^{-1}}$), and produces a measurable density interface at the plume boundary from compositional variation alone. The descending flame therefore introduces two physically distinct density gradient sources simultaneously. The steep thermal gradient of the flame envelope and the broader compositional gradient of the butane-air interface.

Figure 4.21 shows nine RSD frames spanning the propagation event, selected at every 25th frame from a 1000 fps acquisition ($\Delta t = 25$ ms between panels). In the early frames, the candle flame occupies the upper field while the butane plume below appears as a faint green hue variation with a narrow twisted structure at the base. As the flame descends, the flame envelope becomes visible as a warm hue band on its inner face and a cool hue band on its outer face, directly reflecting the sign change of the density gradient across the combustion zone. By the later frames the flame has propagated through the butane plume and the field is dominated by a large coherent warm to cool hue structure. The progression from a quiescent compositional interface to a turbulent flame can be easily observed in the hue field without any postprocessing, demonstrating the interpretive advantage of color schlieren with linear gradient filter for transient flows.

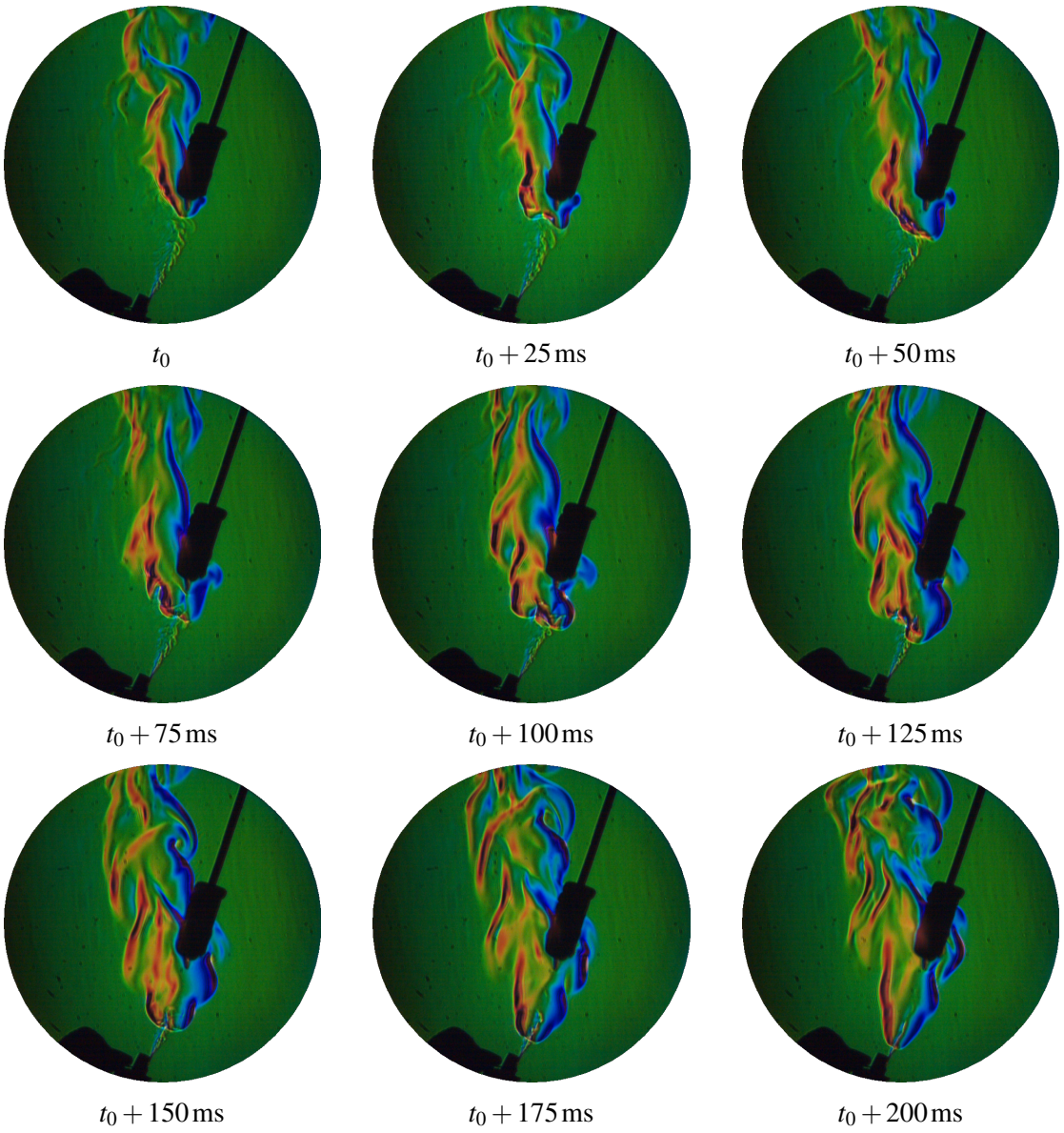


Figure 4.21: RSD image sequence of a candle flame descending into rising butane gas plumes, acquired at 40 fps ($\Delta t = 25$ ms between panels). The green background corresponds to the filter center hue (zero deflection).

4.7 Limitations

i. **Line of sight integration.** RSD integrates over the spanwise depth $L = 150$ mm, yielding a spanwise averaged 2D field. Three dimensional effects near the nozzle side edges are not captured.

ii. **Spatial Resolution.** Due to the limited capacity of zoom lens available, it was not possible to capture finer flow features in detail.

iii. **Light Source Intensity.** The light source available is rated for three watts power and when small aperture is used, the intensity of light is substantially reduced strongly attenuating light in high gradients region.

iv. **density.** The use of 1D linear gradient filter for non axisymmetric flows provides only single gradient field data in one experimental run. For such flows with gradient very small in another direction, one dimensional integration of instantaneous frames produces physically plausible but streaky results.

v. **Velocity measurement.** Available sensors (Pitot ≤ 100 m/s, hot-wire ≤ 35 m/s) could not measure the jet exit velocity ($\approx$ 260–310 m/s). Mach number is estimated from CFD of the plumbing.

vi. **IR thermography.** The polished plate with very low emissivity prevented IRT-based temperature uniformity validation.

vii. **CFD.** 2D RANS does not resolve spanwise or turbulent structures. Supply line loss uncertainty propagates an uncertainty into the inlet Mach number boundary condition.

4.8 Problems Encountered

i. **Supply-line pressure losses.** Not anticipated in the original design. Resolved by CFD simulation of the plumbing to estimate the loss factor ($\approx 30\%$).

ii. **RSD filter fabrication.** Digitally printed filters on transparent sheets were optically inadequate. Slide-film filters procured.

iii. **Schlieren alignment sensitivity.** Z type system required iterative alignment. Minimum achievable Z angle constrained the usable test section width. It was resolved by putting setup in vertical orientation.

iv. **IRT failure.** Specular reflections from the polished plate precluded IR thermography.

v. **Abel inversion on candle plume.** Failed due to plume asymmetry and turbulent transition. Not used in the final analysis.

4.9 Budget Analysis

Table 4.1: Budget analysis.

S.N	Particulars	Amount (NRs)	Remarks
1	Schlieren setup	–	Available in DMAE
2	Compressor–Reservoir	–	Available in DMAE
3	High-speed camera	–	Available in DMAE
4	Nozzle (3D printing)	–	Available in DMAE
5	Rectangular metal plate	500	Procured
6	Peltier modules	2,400	Procured
7	Power supply module	2,000	Procured
8	Pressure regulator	6,000	Procured
9	Arduino Nano microcontroller	1,200	Procured
10	Slide-film colour filter	7,000	Procured
11	Pipes and fittings	500	Procured
12	Other electronics	1000	Procured
13	Miscellaneous	6,000	–
	Total	**26,700**	

Chapter 5: CONCLUSION AND SCOPE OF FURTHER ENHANCEMENT

5.1 Conclusions

The conclusions are drawn from the completed work. Each conclusion corresponds directly to a specific objective stated in Section 1.3.

i. **Facility characterization.** A compressible wall jet facility was designed, built, and characterized. CFD simulation of the full supply plumbing revealed a total-pressure loss of approximately 30% between the regulator outlet and the nozzle inlet. Accounting for Kathmandu's ambient conditions ($p_\infty = 0.871$ bar), the nozzle exits sub-sonically at $M \approx 0.75$ and $M \approx 0.90$ at 1.0 and 1.4 bar gauge respectively, and chokes with an underexpanded supersonic exit at 2.0 bar gauge, confirmed by oblique compression waves in schlieren images.

ii. ***pyrsd* software library.** *pyrsd* (v0.0.1) was developed and documented as an open-source Python library implementing the complete Greenberg RSD pipeline: circular-mean hue calibration, per-frame displacement extraction, ensemble statistics, Gaussian pre-filtering, 1-D cumulative integration, 2-D Poisson inversion, and temperature conversion. The library uses NaN-aware sparse linear algebra and JSON calibration files for reproducibility, and is available at `https://github.com/FVL2082/pyrsd`.

iii. **Ensemble convergence.** Partial means computed at increasing frame counts confirm that the mean displacement field converges to within 1% of its final value by $N \approx 120$ frames.

iv. **Filter and source aperture characterization.** Narrower colour filters (widths 1.5-2.0 mm) provide higher schlieren sensitivity but saturate more readily in high-gradient regions; filters of 2.5–3.0 mm width offer a better balance for the present conditions. The rectangular slit source aperture provides higher schlieren contrast than a circular aperture of comparable area.

v. **Wall-jet density-gradient fields.** Mean density-gradient maps reveal a well-defined high-gradient band at the plate surface corresponding to the coupled shear and thermal

layer. Peak gradient intensity and band width increase monotonically with wall temperature. The 40°C wall case is marginally resolved with the jet on; the 60 and 80°C cases are clearly resolved.

5.2 Scope for Further Enhancement

i. **Direct velocity measurement.** A pressure-rated Pitot tube ($\Delta p > 700$ Pa, corresponding to $V > 310$ m/s) or laser Doppler anemometry would directly validate the estimated nozzle exit, Mach number and provide velocity profiles for CFD comparison.

ii. **High-emissivity plate coating.** Applying a temporary graphite or black-paint coating to a section of the plate would enable IR thermography for spatial temperature uniformity validation, without permanently modifying the experiment.

iii. **Nusselt number for the wall jet.** Increasing the optical magnification to achieve $\Delta x \lesssim 0.05$ mm/px would resolve the near-wall temperature gradient in forced convection and permit Nusselt number extraction for the wall-jet case, enabling direct comparison with heat transfer correlations for compressible wall jets.

iv. **Schlieren imaging velocimetry (SIV).** PIV cross-correlation applied to pairs of high-speed schlieren frames would extract velocity fields from density-gradient structures. At 24,000 fps the inter-frame displacement is ≈46 px at 0.218 mm/px; sub-sampling to every 10th–15th frame would bring displacements into the reliable 1–5 px range for cross-correlation.

v. **Dynamic mode decomposition (DMD).** DMD applied to the same RSD sequences would provide frequency-indexed modes complementing the energy-ranked POD modes, identifying dominant oscillation frequencies in the shear layer.

vi. **3-D CFD with LES.** Large eddy simulation on the DMAE workstation would resolve turbulent structures and enable rigorous POD mode comparison, overcoming the limitations of 2-D RANS.

vii. **Extending *pyrsd*.** Adding SIV cross-correlation, DMD, and background-oriented schlieren (BOS) modules would make *pyrsd* a comprehensive optical flow diagnostics toolkit for the DMAE flow visualization laboratory.

viii. **Abel inversion on a laminar axisymmetric flow.** The Abel inversion path in *pyrsd* should be validated on a clean, stable, laminar axisymmetric flow (e.g., a laminar heated free jet) before application to production measurements.

REFERENCES

[1] J. M. Schulz, H. Junne, L. Böhm, and M. Kraume, “Development of an experimental setup applying rainbow schlieren deflectometry for visualization and quantification of heat and mass transfer in multiphase systems,” in *Proceedings of the 5th International Conference on Experimental Fluid Mechanics* (ICEFM 2018), Munich, Germany, 2018.

[2] P. S. Greenberg, R. B. Klimek, and D. R. Buchele, “Quantitative rainbow schlieren deflectometry,” *Applied Optics*, vol. 34, no. 19, pp. 3810–3825, 1995. doi: 10.1364/AO.34.003810.

[3] A. K. Shukla and A. Dewan, “Flow and thermal characteristics of jet impingement: comprehensive review,” *International Journal of Heat and Technology*, vol. 35, no. 1, pp. 153–166, 2017. doi: 10.18280/ijht.350121.

[4] J. Han and H. Kim, “Pretest of rainbow schlieren deflectometry for measurement of thermal boundary layer thickness,” in *Transactions of the Korean Nuclear Society Autumn Meeting*, 2020.

[5] A. K. Agrawal and C. T. Wanstall, “Rainbow schlieren deflectometry for scalar measurements in fluid flows,” *Journal of Flow Visualization and Image Processing*, vol. 25, no. 3, pp. 329–357, 2018. doi: 10.1615/JFlowVisImageProc.2018028312.

[6] I. A. Joia, R. J. Perkins, B. J. Uscinski, G. Balmer, D. Jordan, and E. Jakeman, “Optical properties of a planar turbulent jet,” *Applied Optics*, vol. 34, no. 30, pp. 7039–7053, 1995. doi: 10.1364/AO.34.007039.

[7] G. E. Elsinga, B. W. van Oudheusden, F. Scarano, and D. W. Watt, “Assessment and application of quantitative schlieren methods: calibrated color schlieren and background-oriented schlieren,” *Experiments in Fluids*, vol. 36, no. 2, pp. 309–325, 2004. doi: 10.1007/s00348-003-0724-8.

[8] R. Goldstein, *Fluid Mechanics Measurements*, 2nd ed. Routledge, 2017.

[9] D. S. Jain, S. S. Rao, and A. Srivastava, “Rainbow schlieren deflectometry technique for nanofluid-based heat transfer measurements under natural convection regime,” *International Journal of Heat and Mass Transfer*, vol. 98, pp. 697–711, 2016. doi: 10.1016/j.ijheatmasstransfer.2016.03.062.

[10] S. Alimohammadi, “A validated computational fluid dynamics methodology for characterisation of fluid flow and heat transfer in unsteady jets,” Ph.D. dissertation, Trinity College Dublin, Dublin, Ireland, 2016.

[11] I. Z. Naqavi, J. C. Tyacke, and P. G. Tucker, “A numerical study of a plane wall jet with heat transfer,” *International Journal of Heat and Fluid Flow*, vol. 63, pp. 99–107, 2017. doi: 10.1016/j.ijheatfluidflow.2016.07.012.

[12] M. B. Glauert, “The wall jet,” *Journal of Fluid Mechanics*, vol. 1, no. 6, pp. 625–643, 1956. doi: 10.1017/S002211205600041X.

[13] P. Bradshaw and M. T. Gee, “Turbulent wall jets with and without an external stream,” Aeronautical Research Council, R&M 3252, 1960.

[14] B. E. Launder and W. Rodi, “The turbulent wall jet: measurements and modelling,” *Annual Review of Fluid Mechanics*, vol. 15, pp. 429–459, 1983. doi: 10.1146/annurev.fl.15.010183.002241.

[15] G. E. Myers, J. J. Schauer, and R. H. Eustis, “Plane turbulent wall jet flow development and friction factor,” *Journal of Basic Engineering*, vol. 85, no. 1, pp. 47–53, 1963. doi: 10.1115/1.3656534.

[16] G. S. Settles, *Schlieren and Shadowgraph Techniques: Visualizing Phenomena in Transparent Media*. Springer-Verlag, Berlin Heidelberg, 2001.

[17] H. Maeda, H. Fukuda, S. Nakao, Y. Miyazato, and Y. Ishino, “Rainbow schlieren measurements in underexpanded jets from square supersonic micro nozzles,” *EPJ Web of Conferences*, vol. 180, p. 02058, 2018. doi: 10.1051/epjconf/201818002058.

[18] G. S. Settles and M. J. Hargather, “A review of recent developments in schlieren and shadowgraph techniques,” *Measurement Science and Technology*, vol. 28, no. 4, p. 042001, 2017. doi: 10.1088/1361-6501/aa5748.

[19] M. J. Hargather and G. S. Settles, “A comparison of three quantitative schlieren techniques,” *Optics and Lasers in Engineering*, vol. 50, no. 1, pp. 8–17, 2012. doi: 10.1016/j.optlaseng.2011.05.012.

[20] C. Lucio Cifuentes, “Design of a high subsonic nozzle for a transonic linear cascade operating at near stall conditions,” Bachelor’s thesis, KTH School of Industrial Engineering and Management, Stockholm, Sweden, TRITA-ITM-EX 2021:268, 2021.

[21] W. H. Rae and A. Pope, *Low-Speed Wind Tunnel Testing*, 2nd ed. Wiley, 1984.

[22] A. Adhikari, A. Bajgain, T. B. Niroula, and Y. Dhami, “pyrsd: A Python library for rainbow schlieren deflectometry processing,” [Software], version 0.1.0-alpha, 2026. doi: 10.5281/zenodo.19219233.

[23] A. K. Agrawal, K. Alammar, and S. R. Gollahalli, “Application of rainbow schlieren deflectometry to measure temperature and oxygen concentration in a laminar gas-jet diffusion flame,” *Experiments in Fluids*, vol. 32, no. 6, pp. 689–691, 2002. doi: 10.1007/s00348-002-0422-y.

[24] T.-H. Shih, W. W. Liou, A. Shabbir, Z. Yang, and J. Zhu, “A new k-ε eddy viscosity model for high Reynolds number turbulent flows: model development and validation,” *Computers & Fluids*, vol. 24, no. 3, pp. 227–238, 1995.

[25] F. R. Menter, “Two-equation eddy-viscosity turbulence models for engineering applications,” *AIAA Journal*, vol. 32, no. 8, pp. 1598–1605, 1994. doi: 10.2514/3.12149.

[26] F. R. Menter, M. Kuntz, and R. Langtry, “Ten years of industrial experience with the SST turbulence model,” in *Turbulence, Heat and Mass Transfer 4*, pp. 625–632, Begell House, 2003.

[27] D. C. Wilcox, *Turbulence Modeling for CFD*, 3rd ed. DCW Industries, 2006.

[28] S. Sarkar, G. Erlebacher, M. Y. Hussaini, and H. O. Kreiss, “The analysis and modelling of dilatational terms in compressible turbulence,” *Journal of Fluid Mechanics*, vol. 227, pp. 473–493, 1991. doi: 10.1017/S0022112091000204.

[29] A. Dejoan and M. A. Leschziner, “Large eddy simulation of a plane turbulent wall jet,” *Physics of Fluids*, vol. 17, no. 2, p. 025102, 2005. doi: 10.1063/1.1833413.

Chapter A: NOZZLE DESIGN: QUASI-ONE-DIMENSIONAL ANALYSIS

The nozzle considered is a three-dimensional rectangular contraction, convergent in both the height (x) and width (y) directions, with streamwise coordinate z. The cross-sectional area is $A(z) = H(z)\,W(z)$.

Assuming steady, inviscid, isentropic flow, the local Mach number $M(z)$ is related to the local area through the isentropic area–Mach relation:

$$\frac{A(z)}{A^*} = \frac{1}{M(z)}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2(z)\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}. \tag{A.1}$$

Local static quantities follow from isentropic relations:

$$\frac{T(z)}{T_0} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-1}, \quad \frac{p(z)}{p_0} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-\gamma/(\gamma-1)}, \quad \rho(z) = \frac{p(z)}{RT(z)}. \tag{A.2}$$

The height profile follows a 5th-order polynomial, ensuring zero slope and curvature at inlet ($z = 0$) and exit ($z = L$):

$$H(z) = H_1 - (H_1 - H_2)\left[6\left(\tfrac{z}{L}\right)^5 - 15\left(\tfrac{z}{L}\right)^4 + 10\left(\tfrac{z}{L}\right)^3\right]. \tag{A.3}$$

Width contracts linearly: $W(z) = W_1 - (W_1 - W_2)z/L$.

The nozzle exit (throat for a converging nozzle) chokes when $\dot{m} = \dot{m}_{\max}$, where

$$\dot{m}_{\max} = \sqrt{\gamma p_0 \rho_0}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} A_{\text{exit}}. \tag{A.4}$$

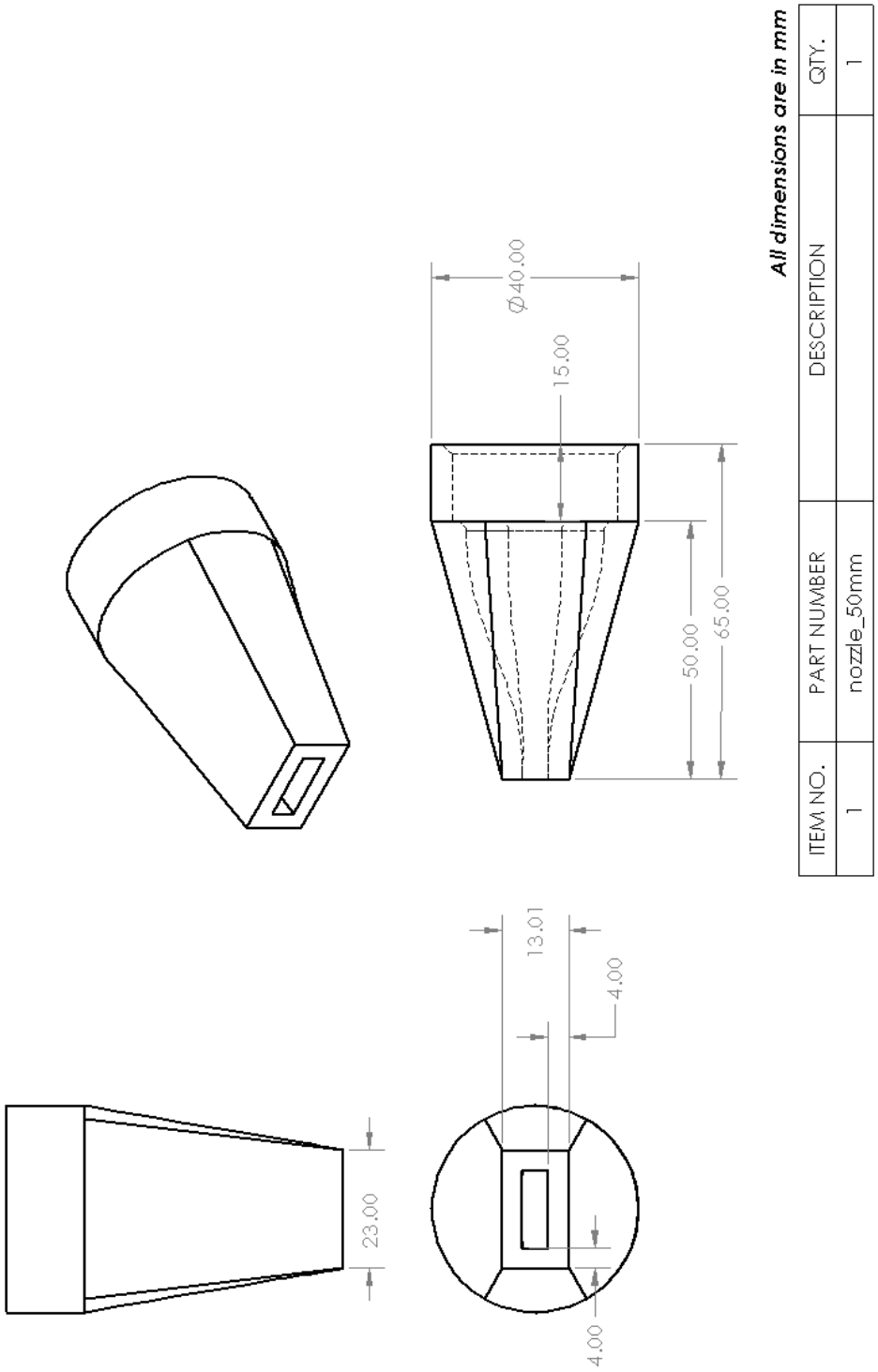


Figure A.1: Engineering drawings of the 3D-printed converging nozzle.